When Can a Machine Judge Like a Human?

Evaluating LLM-as-a-Judge Approaches for Human-Centered Computing Research

Anonymous Author(s)*

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Abstract

Large language models (LLMs) are flooding into Human-Centered Computing (HCC) research as substitutes for human judgment: coding qualitative data, annotating crowdsourced corpora, simulating study participants, generating personas, and evaluating designs. Yet HCC's methods carry epistemological commitments (interpretation, reflexivity, positionality) that make uncritical substitution risky, and the field lacks an evidence-based account of *when* an "LLM-as-a-judge" can stand in for a human and when it cannot. We address this with four studies. **Study 1a** is a PRISMA systematic review of 215 studies across 29 domains that establishes the general reliability landscape of LLM-as-a-judge: substitution is validated ($\approx$ or above the human-human ceiling) for objective, verifiable, low-expertise tasks judged in aggregate, and unreliable for subjective, expert, low-resource, or adversarial ones (mean reliability 3.15/5; only 31% of studies in the reliable band). **Study 1b** asks whether this landscape is a moving target: we re-run a flagship "promising" benchmark (SummEval) with the most powerful open models served via an institutional vLLM API and ask whether they now *surpass* the agreement numbers reported for proprietary judges, distinguishing a "matter of time" from a structural ceiling. **Study 2** is a PRISMA systematic review of 55 studies focused on HCC, HCI, and qualitative-research methods that historically require human input. It finds that the field overwhelmingly positions LLMs to *augment* rather than replace humans (36 augment vs. 14 replace vs. 5 simulate), with an epistemic stance dominated by *tension* (38/55); objective crowd-annotation substitution is validated, interpretive qualitative analysis is reliable only as human-in-the-loop augmentation, and *simulating* participants, personas, and usability evaluations is the least reliable (flattening of identity groups, caricature, hallucinated usability issues). **Study 3** closes the gaps the first two studies expose, by running a standardized LLM-as-a-judge meta-evaluation, using open models served via an institutional vLLM API and the agreement-against-human-ceiling methodology distilled from Studies 1-2, on HCC datasets that contain human labels but have never been formally tested with an LLM judge. Across 18000 judgments (a 6-model open panel, 9B-754B), it finds the human-agreement ceiling governs reliability: the panel matches or exceeds humans on low-ceiling NLI but falls short of, and fails the alt-test for, subjective social-computing annotation (emotion, hate speech), with binary offensiveness the strongest case; a recent low-contamination irony set is matched only at its own low human ceiling, which (with an independent contamination control) argues against memorization as the explanation. An inferential analysis localizes the mechanism: judge correctness is governed by *item-level human (dis)agreement* (random-intercept $SD_{\text{item}} = 2.02$, far exceeding dataset- and model-level variance) rather than by a clean dataset-level "subjectivity" term (which is not robust once label base-rate and granularity are controlled); judges lose accuracy precisely on high-disagreement items, and most judge$\times$dataset cells sit significantly below the human ceiling. An independent re-coding

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of the review corpora gives Krippendorff $\alpha = 0.87$ on the reliability verdict. By design the empirical studies use only *open, locally-servable* models, the realistic choice for most researchers, and ground the annotation regime empirically while the interpretive regimes (qualitative coding, participant simulation, design judgment) are synthesized from the review. Together the studies yield a decision framework telling HCC researchers when an LLM judge is a validated substitute, a human-in-the-loop assistant, or an unsafe replacement.

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1 Introduction and motivation

In barely two years, LLMs have moved from objects of HCI study to *instruments* of HCI method. Researchers now use them to code interview transcripts [22, 55], run thematic analyses [16, 43], annotate social-media corpora [23, 50], simulate survey respondents and "silicon samples" [2], generate personas [45], and even perform heuristic usability evaluation [25]. The appeal is obvious: human judgment is the costly bottleneck of human-centered research, and an LLM that labels for $0.003 an item is a tempting stand-in [23].

The danger is equally clear. HCC and qualitative methods are not mere labeling pipelines; they carry *epistemological commitments* (interpretation, reflexivity, positionality, and the situated knowledge of the researcher) that an LLM cannot obviously honor [17, 47]. Substituting a model for a human participant can *flatten* and misportray the very identity groups HCI seeks to center [14, 53]. And the empirical record is genuinely mixed: the same model can match two human coders on a structured codebook [19] yet hallucinate usability problems that do not exist [25], or agree with a second annotator on stance [23] yet be fooled into mislabeling relevance by a few injected keywords [1].

HCC researchers therefore face a question with no consolidated answer: **for which HCC research tasks is an LLM-as-a-judge a safe substitute for human judgment, for which is it a useful assistant, and for which is it unsafe?** This paper answers that question through three studies with a deliberate progression (Figure 1):

1. **Study 1a, the general landscape.** A PRISMA systematic review of 215 studies across 29 domains establishes *where, in general*, LLM-as-a-judge agrees with humans, anchored to the human-human ceiling. This gives HCC the broad map and the methodological vocabulary (the reliability rubric, the bias catalogue, the decision map).
2. **Study 1b, is the landscape a moving target?** The review captures results *as reported*, often with one- or two-generation-old proprietary judges. We re-run a flagship "promising"-band benchmark (SummEval summarization quality) with the *most powerful open models available today* and ask whether they now surpass the canonical reported judge correlations, separating limitations that are a "matter of time" (scale/recency) from *structural* ceilings that more capable models do not cross.
3. **Study 2, the HCC lens.** A second PRISMA review of 55 studies zooms into the methods HCC actually uses (qualitative coding, thematic/content analysis, crowdsourced annotation,

Study 1a: *Where* is LLM-as-a-judge reliable? (cross-domain review, 215 studies) → **Study 1b**: *Is it a moving target?* (most powerful open models vs reported numbers, SummEval) → **Study 2**: *How does HCC use LLMs* for human-input tasks? (55 studies) → **Study 3**: *Fill the gaps*, standardized evaluation on HCC datasets with human labels (open models via an institutional vLLM API).

Figure 1: The four-study progression. Study 1a maps where LLM judges are reliable; Study 1b tests whether today's strongest open models move that map; Study 2 narrows to Human-Centered Computing practice; Study 3 supplies new empirical evidence on under-tested HCC datasets.

participant simulation, persona generation, usability evaluation, and design critique) and codes how the field positions LLMs (augment / replace / simulate) and with what epistemic stance.

4. **Study 3, filling the gaps.** Studies 1-2 reveal HCC datasets that contain human labels but have never been formally evaluated with an LLM judge. Study 3 runs a standardized LLM-as-a-judge meta-evaluation on these datasets, using open models served through an institutional vLLM API and the agreement-against-human-ceiling methodology distilled from the prior literature, with a mixed-effects analysis of what governs judge reliability.

Contributions. (1) The first consolidated, human-ceiling-anchored map of LLM-as-a-judge reliability across 29 domains (Study 1a). (2) A direct test of whether the most powerful open models available today surpass the reported judge correlations in a flagship "promising" domain, adjudicating "matter of time" vs. structural ceiling (Study 1b). (3) The first systematic review of LLM substitution *specifically* for HCC/HCI/qualitative methods, with a method×role×stance coding that surfaces the field's augment-not-replace consensus and its epistemic tensions (Study 2). (4) A reproducible, standardized evaluation harness plus a mixed-effects account of what governs judge reliability (Study 3), tested on HCC datasets that had human labels but no prior LLM-judge evaluation, plus two released corpora (215 + 55 coded studies), all code, and a decision framework for practitioners.

2 Background and shared methodology

LLM-as-a-judge descends from reference-free metrics (BERTScore, COMET) but uses a general instruction-following model with a natural-language rubric, supporting pointwise, pairwise, and listwise judgment [24, 33]. Four architectural families recur: prompted general LLMs [36, 57]; fine-tuned open evaluators [31, 52, 59]; juries/multi-agent panels [13, 51]; and calibrated rubric methods [26]. Prior surveys taxonomize methodology [24, 34]; we instead synthesize *where substitution is empirically validated*, and, uniquely, do so through the HCC lens.

Related work. Four literatures meet in this paper. (1) *LLM-as-a-judge evaluation*: studies report strong agreement on aggregate, verifiable tasks (MT-Bench [57], GEMBA [32], SAFE factuality [54]) but document systematic biases (position, verbosity, self-preference, and an expertise gap by which judges converge on crowd rather than expert consensus [4]), which motivates benchmarking against the human ceiling rather than against a single "gold." (2) *LLMs in HCI and qualitative research*: a fast-growing body uses LLMs to assist coding and thematic analysis [16, 22, 55] while qualitative scholars warn that interpretation, reflexivity, and positionality resist automation [17, 47]. (3) *Participant and persona simulation*: "silicon samples" [2] and synthetic users are increasingly scrutinized for flattening identity and distorting variance [7, 14, 53]. (4) *Learning with disagreement and perspectivism*: work on annotator disagreement and human label variation [9] argues that for subjective constructs the "ceiling" is itself low and contested, which our Study 3 datasets (ChaosNLI, GoEmotions, MultiPICo) operationalize directly. We connect these threads with two methodological commitments often missing from judge evaluations: anchoring every verdict to the *human-human* ceiling, and probing *data contamination* so that agreement on older public datasets is not mistaken for genuine competence.

Shared protocol. Both reviews follow PRISMA 2020 [41], identify studies through a triangulated search (the OpenAlex API as the primary bibliographic database, Semantic Scholar as a cross-check, and a structured web "deep-research" arm), and code each study on a five-level **reliability verdict** benchmarked against the *human-human agreement ceiling* for the same task: **1 Validated** ($\approx$/> ceiling), **2 Promising** (conditional), **3 Mixed** (task-dependent), **4 Caution** (below ceiling), **5 Unreliable** (near chance / bias-dominated). We report a reliability score = 6−verdict and synthesize within metric families rather than pooling incommensurable effect sizes. Full protocols, search logs, and the two coded corpora are released.

3 Study 1a: The cross-domain reliability landscape

3.1 Method (summary)

Study 1a systematically reviewed 215 studies across 29 domains comparing LLM-as-a-judge to human annotation. Identification triangulated OpenAlex (1500 records, 957 unique), Semantic Scholar, and a 59-query web search; screening retained on-topic empirical comparisons; 29 studies reported an extractable numeric agreement value (Figure 2; the full coded corpus is Table 7). **Coding reliability:** because the verdict rubric embeds judgement, we validated it with an independent re-coding of a 25% random sample (n=54): the reliability verdict reproduced at Krippendorff $\alpha = 0.87$ (ordinal) with 100% of cases within one level (Appendix D). The re-coders are LLM-assisted, like the primary pipeline, so this measures rubric *reproducibility* rather than human inter-rater agreement, which we flag as a limitation. **Citation integrity:** every one of the 215+55 references was independently web-verified for existence against arXiv/venue/DOI records; 270/272 were confirmed (260 at high confidence) and the 2 that could not be verified were removed, so the corpus contains no unverifiable citations.

3.2 Findings: a green/amber/red landscape

The modal verdict is *mixed/task-dependent*; only 31% of studies sit in the reliable band and 19% in the problematic band (mean reliability 3.15/5). Reliability is governed by three properties: **objectivity**, **tractable verification**, and a **low human-agreement floor**. **Green ($\geq$3.5; 8 domains):** machine translation (system level), text-to-image alignment, verifiable instruction-following, QA/factuality, information extraction, general-NLG pairwise preference, dialogue, and legal scoring: here LLM judges frequently match or exceed the human ceiling (e.g., MT-Bench 85% vs. 81% human-human [57]; GEMBA-MQM 96.5% [32]; SAFE wins 76% of disagreements [54]). **Amber (19 domains):** summarization, RAG, education, social-science annotation, medicine (structure-dependent: $\kappa \approx$0.82-0.93 structured vs. 0.47-0.71 complex reasoning), code, and IR/relevance (strong but contested [15]). **Red (2 domains):** low-resource multilingual evaluation and creative writing (near-zero expert correlation [12]). Figures 3a-3b give the domain ordering and the domain×task landscape; the decision map (Figure 4b) positions domains by reliability against stakes.

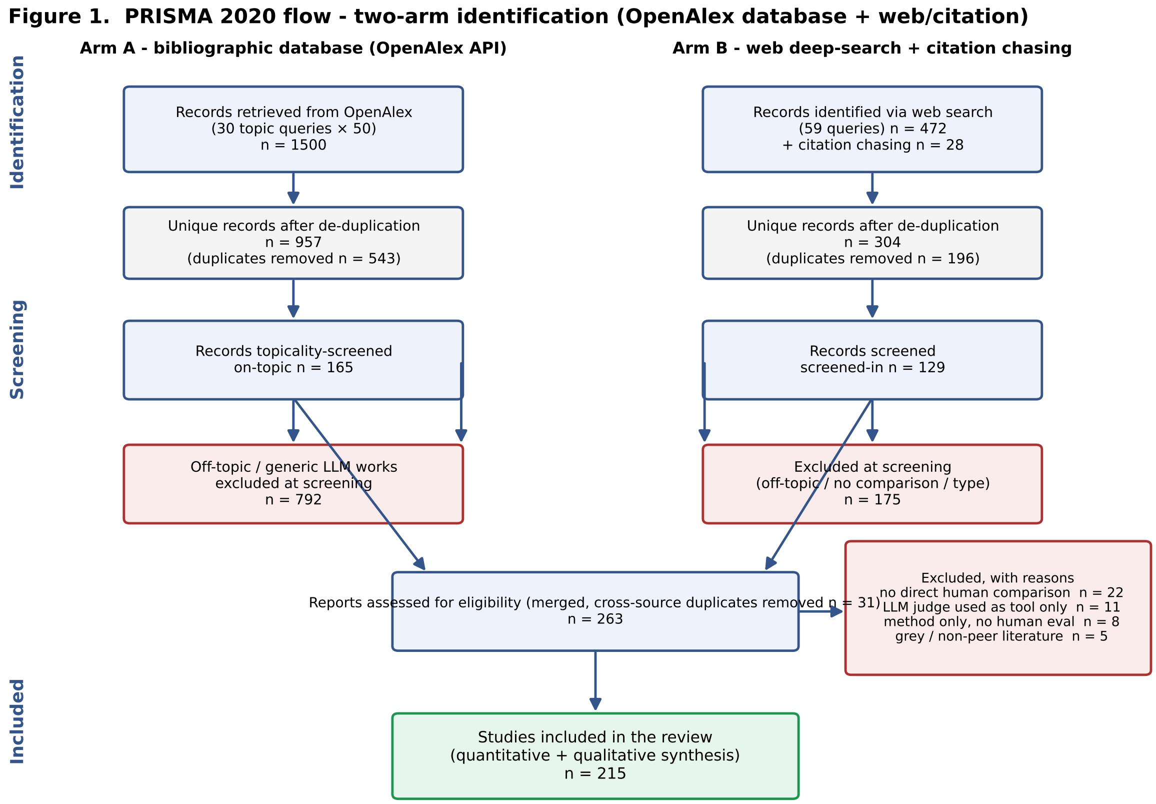


Figure 2: Study 1a PRISMA 2020 flow (two-arm identification: OpenAlex database + web/citation).

3.3 Failure modes and the decision map

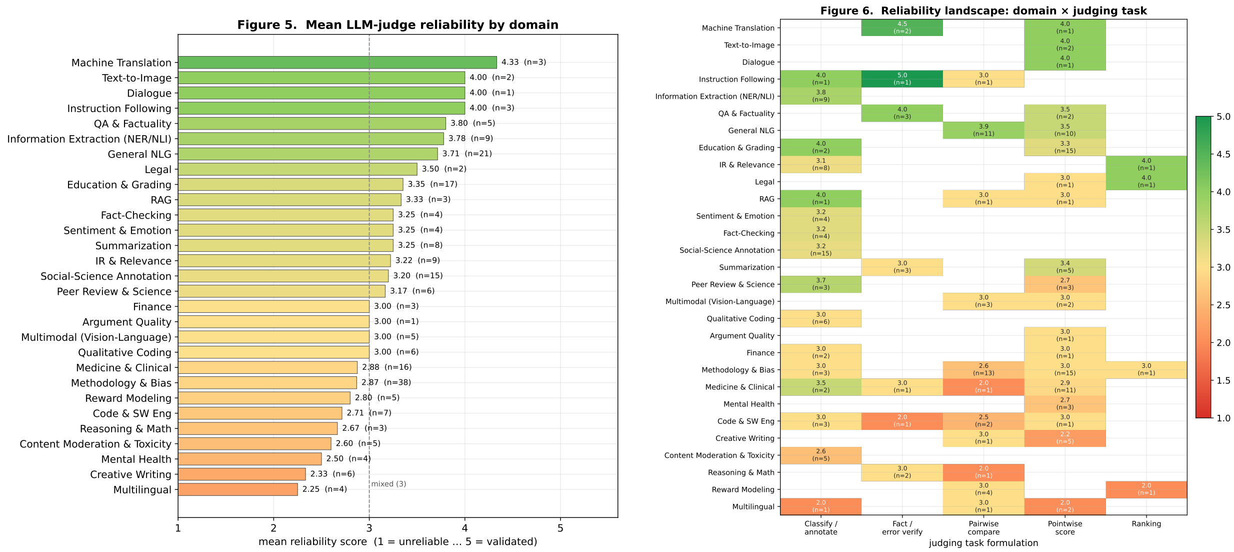
Recurring biases such as verbosity/length, position, self-preference, style-over-substance, preference leakage, non-determinism, and the *expertise gap* (judges converge to crowd, not expert, consensus [4]) all modulate reliability (Figure 4a). Agreement should be benchmarked against the human ceiling, which on subjective tasks is itself low (Krippendorff $\alpha < 0.35$ [30]). The Study-1a decision map (Figure 4b) and Table 1 translate this into deployment guidance and motivate both the moving-target test (Study 1b) and the HCC-specific analysis that follows.

4 Study 1b: Is the reliability landscape a moving target?

A systematic review necessarily captures results *as reported*, and most "promising"-band evidence in Study 1a was produced with proprietary judges that are now one or two model generations old (e.g. the influential G-Eval used GPT-4). This raises a question that the review alone cannot answer: when an LLM judge falls short of the human ceiling in a promising domain, is that a *temporary* limitation that more capable models will erase ("matter of time"), or a *structural* property of LLM-as-a-judge that better models do not overcome? Study 1b answers this directly for a flagship case by re-running the evaluation with the most powerful open models available to us today and comparing against the canonical reported numbers.

Table 1: Domain-level decision table. Mean reliability score (1 = unreliable ... 5 = validated), modal verdict, and traffic-light signal (GREEN ≥ 3.5; AMBER 2.5-3.5; RED < 2.5). Sorted by mean reliability. Small-n domains should be read together with the family-level aggregates.

Domain / task	n	Mean rel.	Modal verdict	Signal
Machine Translation	3	4.33	Promising	GREEN
Text-to-Image	2	4.00	Promising	GREEN
Dialogue	1	4.00	Promising	GREEN
Instruction Following	3	4.00	Validated	GREEN
QA & Factuality	5	3.80	Promising	GREEN
Information Extraction (NER/NLI)	9	3.78	Promising	GREEN
General NLG	21	3.71	Promising	GREEN
Legal	2	3.50	Promising	GREEN
Education & Grading	17	3.35	Mixed	AMBER
RAG	3	3.33	Mixed	AMBER
Summarization	8	3.25	Promising	AMBER
Fact-Checking	4	3.25	Promising	AMBER
Sentiment & Emotion	4	3.25	Mixed	AMBER
IR & Relevance	9	3.22	Promising	AMBER
Social-Science Annotation	15	3.20	Mixed	AMBER
Peer Review & Science	6	3.17	Mixed	AMBER
Multimodal (Vision-Language)	5	3.00	Mixed	AMBER
Qualitative Coding	6	3.00	Mixed	AMBER
Argument Quality	1	3.00	Mixed	AMBER
Finance	3	3.00	Mixed	AMBER
Medicine & Clinical	16	2.88	Mixed	AMBER
Methodology & Bias	38	2.87	Mixed	AMBER
Reward Modeling	5	2.80	Mixed	AMBER
Code & SW Eng	7	2.71	Mixed	AMBER
Reasoning & Math	3	2.67	Mixed	AMBER
Content Moderation & Toxicity	5	2.60	Mixed	AMBER
Mental Health	4	2.50	Mixed	AMBER
Creative Writing	6	2.33	Mixed	RED
Multilingual	4	2.25	Caution	RED



(a) (b)

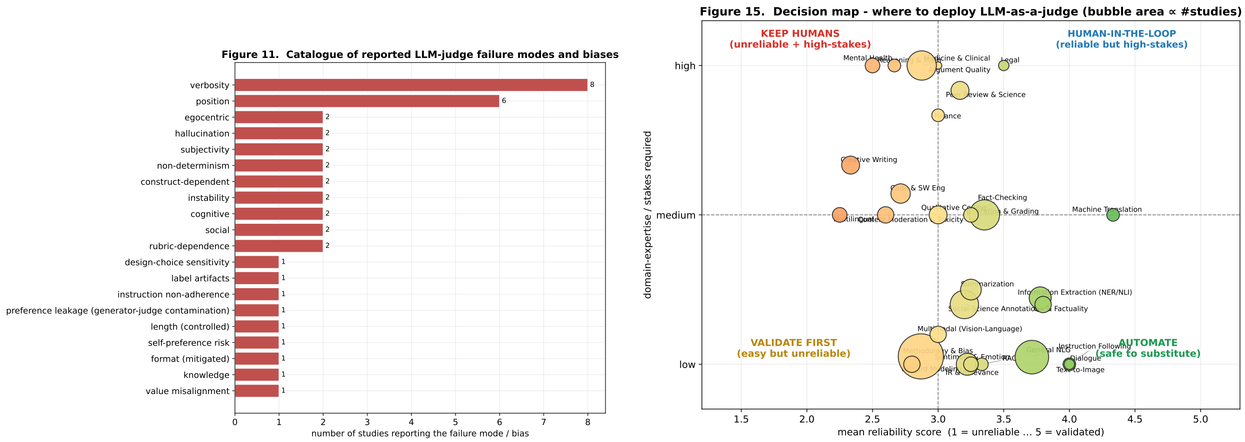
Figure 3: Study 1a. (a) Mean reliability by domain (green/amber/red). (b) Reliability landscape: domain×judging task; verification/classification columns are greener than open-ended scoring.

Table 2: Study 1b. Summary-level Spearman ρ of the most powerful open judges on SummEval vs. the reported G-Eval-4/GPT-4 correlations (25,600 judgments). Green cells *surpass* the reported per-aspect number.

Judge	Coher.	Consist.	Fluency	Relev.	Avg ρ
Reported (G-Eval-4)	0.582	0.507	0.455	0.547	0.523
`deepseek-v4-pro`	0.548	0.688	0.392	0.540	0.542
`qwen3.6-35b-a3b`	0.606	0.702	0.298	0.540	0.536
`glm-5.1`	0.650	0.693	0.217	0.543	0.526
`gptoss-120b`	0.586	0.685	0.260	0.482	0.503

4.1 Method

We select **SummEval** [20] (summarization quality assessment, a central amber-band task in Study 1a) because it pairs (i) a widely cited LLM-judge result to beat (G-Eval/GPT-4; 36) with (ii) public expert human ratings on four aspects (coherence, consistency, fluency, relevance) for $100 \times 16 = 1600$ machine summaries. We replicate the G-Eval summary-level protocol *verbatim* (same aspect definitions, 1-5 integer ratings, temperature 0) but swap the judge for the 4 most powerful open models in the available open-model catalogue (`deepseek-v4-pro`, `glm-5.1`, `gptoss-120b`, `qwen3.6-35b-a3b`). The comparison metric is the canonical *summary-level Spearman*: for each source document, correlate the judge's 16 scores with the human means, averaged across documents. The bar to beat is G-Eval/GPT-4's reported per-aspect correlations (coherence .582, consistency .507, fluency .455, relevance .547; average ≈.52). In total Study 1b contributes 25,600 fresh judgments.



(a) (b)

Figure 4: Study 1a. (a) Catalogue of reported failure modes/biases. (b) Deployment decision map (reliability× stakes; bubble area ∝ studies).

4.2 Results

We follow the standard summary-level G-Eval protocol [36] and attach bootstrap 95% CIs (over the 100 source documents) to every correlation (Figure 5, Table 2). On the verifiable aspects the open judges *meet or exceed* the established G-Eval/GPT-4 benchmark: 5 of the 16 model×aspect cells have a 95% CI lying *entirely above* the reported value, all on **consistency** and **coherence** (e.g. `deepseek-v4-pro` consistency ρ=0.69, CI $[0.63, 0.74]$ vs. .507), and **relevance** is on par (.48-.54 vs. .55). Our protocol differs from the original in two deliberate, deployment-relevant ways: the judges are open and locally-servable, and we read a single integer score rather than a proprietary model's probability-weighted decoding. The comparison is to the figures reported by the original work, the standard reference point in this literature. But on **fluency** every open model falls well *short* (.22-.39 vs. a reported .455). We diagnose why: fluency is the most *compressed* aspect of the human reference, with the lowest within-document rating variance (SD = 0.63 vs 0.81 for consistency) and 80% of summaries rated near the top of the scale, so it offers the least to rank, and it is the aspect on which the *reported proprietary judge also scores lowest*. This is a property of the *reference*, not the judges: even a noiseless judge returning the rounded human means tops out at ρ≈0.87 on fluency. The clean result is therefore that capability gains concentrate on the verifiable, rankable aspects (consistency, coherence), while the most compressed aspect is hard to rank for every judge, open or proprietary.

Interpretation: time vs. structure. Study 1b shows the reliability landscape is *partly* a moving target. Where Study 1a's amber rating reflected a capability gap on a verifiable sub-judgment, today's open models close or exceed it; reliability there is largely a matter of time and access. But where the shortfall reflects a *compressed or contested human ceiling* (e.g. fluency, and, as Study 3 will show, subjective social annotation), even the most powerful models do not break through, because the limit lives in the human reference, not the model. This distinction, temporal capability gap vs. structural ceiling, is the lens we carry into HCC.

Takeaway for HCC. Study 1a says LLM judges are safe where the task is objective and verifiable, and Study 1b shows that frontier open models are now strong enough to match proprietary judges on exactly those verifiable tasks. But HCC's signature methods (qualitative interpretation, participant

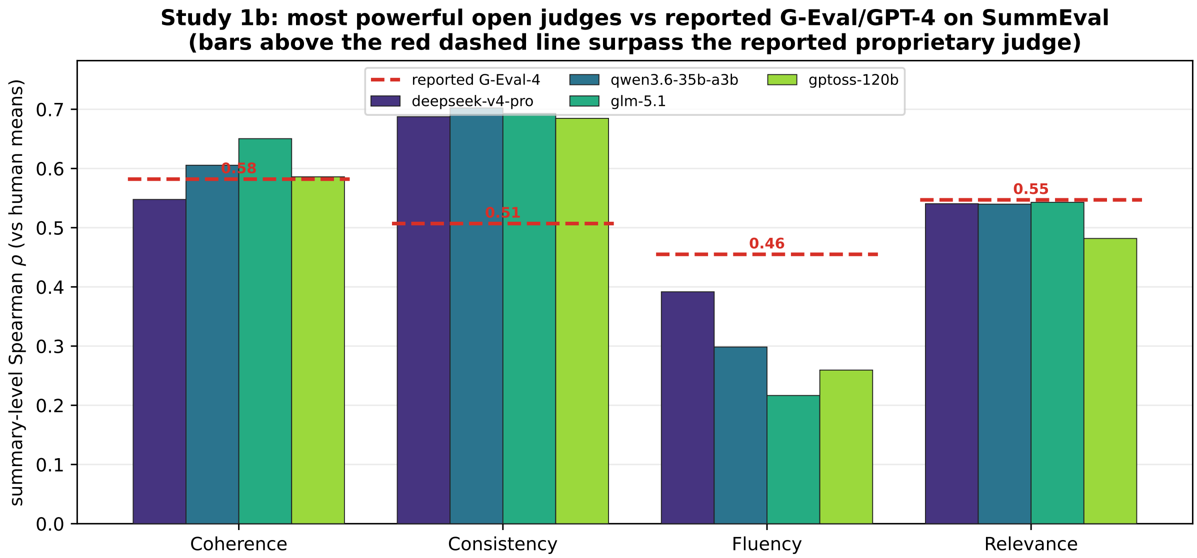


Figure 5: Study 1b. Summary-level Spearman of the most powerful open judges on SummEval, per aspect, against the reported G-Eval/GPT-4 bar (dashed). Bars above the dashed line indicate an open model that *surpasses* the reported proprietary-judge correlation.

study, design judgment) are precisely the *subjective, expert, interpretive* tasks Study 1a flags as risky and Study 1b predicts scale will *not* fix. Study 2 tests whether that prediction holds in the HCC literature itself.

5 Study 2: LLM substitution in HCC, HCI, and qualitative methods

5.1 Motivation and research questions

Study 1's map is domain-general; HCC needs a method-specific account. Human-centered research is built on methods that historically *require* human input: qualitative coding, thematic and content analysis, interviews and open-ended survey analysis, crowdsourced annotation, participant studies, persona development, usability/heuristic evaluation, and design critique. Study 2 asks: **(RQ2.1)** which of these methods have been attempted with LLMs, and how reliably do LLM outputs agree with human input? **(RQ2.2)** How does the field position the LLM, to *augment*, *replace*, or *simulate* the human? **(RQ2.3)** What epistemic stance (enabling, tension, critical) does the literature take, and what does that imply for HCC practice?

5.2 Method

Following the shared protocol, we ran a Study-2-specific identification arm: 25 HCC/HCI/qualitative concept-block queries against OpenAlex (857 unique works), triangulated with the web arm and with the venue-scoped sweep of Study 1 (CHI, CSCW, UIST, IUI, TOCHI). After topicality screening and eligibility assessment we included 55 studies (Figure 6). An independent re-coding of a 30% sample (n=16) reproduced the verdict at Krippendorff $\alpha = 0.85$ and the augment/replace/simulate role at $\alpha = 0.80$; the *epistemic stance* code is the least reliable ($\alpha = 0.59$), so we treat stance-based claims as indicative rather than precise (Appendix D). We extracted, per study: the HCC **method**; the LLM **role** (replace / augment / simulate); reported agreement and the human baseline; the

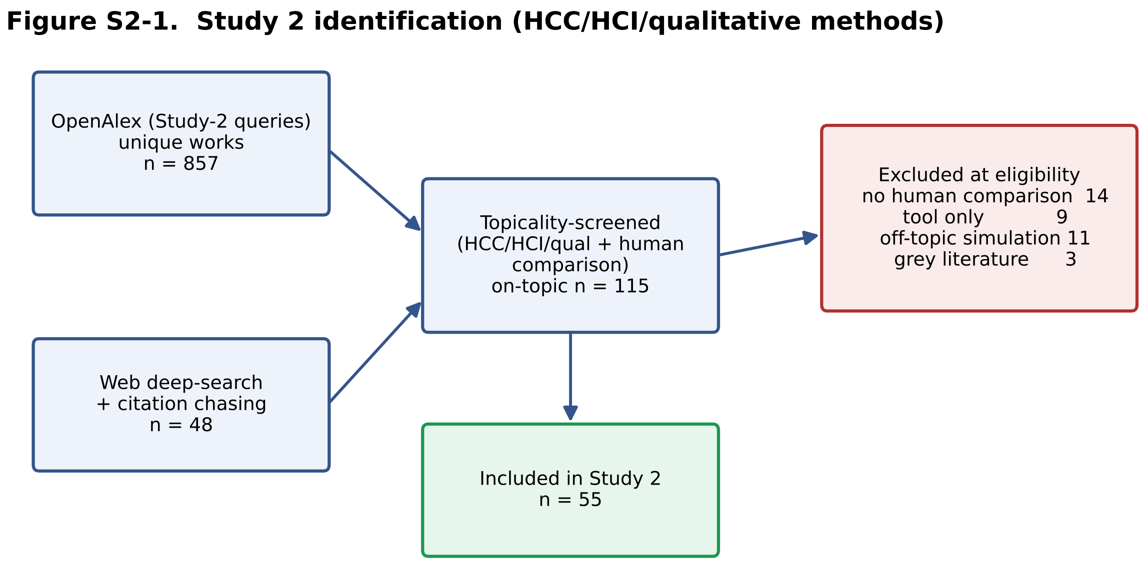


Figure 6: Study 2 identification flow (HCC/HCI/qualitative methods).

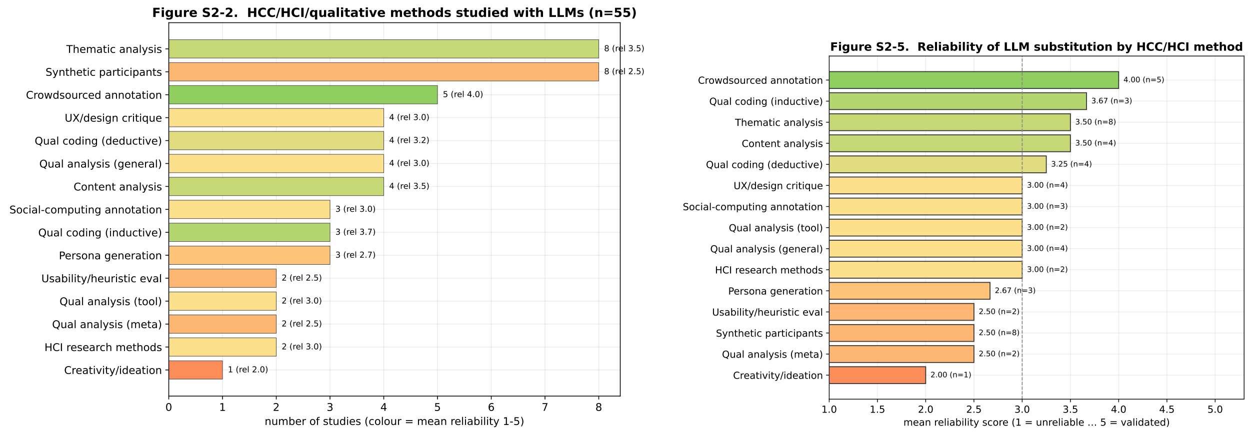
reliability **verdict**; the **epistemic stance** (enabling / tension / critical); and the **community** (HCI, CSCW, qualitative-methods, computational social science, NLP, health). The full corpus is Table 8.

5.3 RQ2.1: Which methods, and how reliable?

LLMs have been compared against human input across 15 distinct HCC methods (Figure 7a), most heavily in *thematic analysis*, *synthetic participants*, and *crowdsourced annotation*. Reliability is strongly method-dependent (Figure 7b; Table 3). The *same objectivity/interpretation axis* from Study 1 governs HCC: **objective crowd-annotation** is the most reliable method to substitute: ChatGPT matches or beats crowd workers and even trained coders on stance, topic, and frame labeling [23, 50]; **deductive/inductive coding and content/thematic analysis** are reliable *as augmentation*, with human-equivalent agreement on structured codebooks (GPT-4 exceeding substantial κ on 8/9 hermeneutic tasks [19]; Gemini $\kappa = 0.907$ in multi-LLM thematic analysis [29]; 79.7% agreement in health thematic summarization [11]) but only moderate agreement in open interpretive settings ($\kappa \approx 0.40$-0.46 [35, 56]); and **participant/persona simulation and usability evaluation** are the least reliable.

5.4 RQ2.2: Augment, replace, or simulate?

HCC overwhelmingly positions LLMs to *augment* humans, not replace them: 36 of 55 studies frame the LLM as a collaborator/assistant, versus 14 as a replacement and 5 as a human simulator (Figure 8a). This is not merely rhetorical caution; it tracks reliability (Figure 9): augmentation studies skew reliable, whereas *replacement* is validated only for objective annotation and *simulation* of participants is the least reliable role (mean reliability augment 3.17 vs. replace 3.07 vs. simulate 2.8). The replacement mean is buoyed by objective crowd-annotation; remove it and replacement of *interpretive* human input is firmly amber/red. The canonical HCC workflow that emerges is human-in-the-loop: LLM proposes, human vets and retains interpretive authority [16, 22, 38].



(a) (b)

Figure 7: Study 2. (a) HCC/HCI/qualitative methods studied with LLMs (colour = mean reliability). (b) Mean reliability of LLM substitution by method.

5.5 RQ2.3: Epistemic stance and the limits of substitution

The dominant stance is *tension* (38/55), with 8 explicitly *critical* and only 9 unreservedly *enabling* (Figure 8b). Three threads recur. **(i) Interpretation and reflexivity.** Qualitative scholars warn that LLMs perform surface-level extraction well but cannot supply the situated, reflexive interpretation that gives qualitative findings their validity, and that "conditional trust" in tool output risks hollowing out analytic rigor [17, 39, 40, 47]. **(ii) Participants and identity.** Replacing human participants with LLMs *flattens and misportrays* demographic groups, positionality cannot be simulated [53], and LLM "subgroups" are often caricatures [14]; synthetic survey data distort variances and correlations [7], and an LLM is worth only a bounded number of real respondents [28]. HCI critiques accordingly recast synthetic users as "objects of inquiry," not stand-ins [27, 46]. **(iii) Expert design judgment.** GPT-4o overlaps only 21% with expert-found usability issues and hallucinates false positives [25], and LLM UI critique trails expert designers [18], though synthetic heuristic evaluation can exceed *novice* evaluators on issue recall [58]. Figure 10 shows coverage skews to HCI and computational social science, with the augment-replace reliability gap visible across communities.

5.6 Study 2 synthesis

For HCC research, the evidence supports a clear, conditional posture. **Validated to substitute:** objective crowd/data annotation (stance, topic, relevance, frames) where a human ceiling is modest and verifiable. **Reliable only as augmentation (human-in-the-loop):** deductive/inductive qualitative coding and thematic/ content analysis: LLMs accelerate breadth and first-pass coding, but humans must own interpretation, theme development, and reflexive accounting. **Unsafe to substitute:** simulating human participants, generating personas as stand-ins for real users, and replacing expert usability/design judgment: these flatten identity, fabricate issues, and forfeit the situated knowledge that defines human-centered inquiry. This posture both confirms Study 1's objective/subjective prediction *within* HCC and exposes concrete gaps (§6) where HCC datasets with human labels have never been formally tested with an LLM judge.

Table 3: Study 2 method-level decision table: study count, modal LLM role, mean reliability, and traffic-light signal (GREEN ≥ 3.5; AMBER 2.5-3.5; RED < 2.5).

Method	n	Modal role	Mean rel.	Signal
Crowdsourced annotation	5	replace	4.00	GREEN
Qual coding (inductive)	3	augment	3.67	GREEN
Content analysis	4	augment	3.50	GREEN
Thematic analysis	8	augment	3.50	GREEN
Qual coding (deductive)	4	augment	3.25	AMBER
HCI research methods	2	augment	3.00	AMBER
Qual analysis (general)	4	augment	3.00	AMBER
Qual analysis (tool)	2	augment	3.00	AMBER
Social-computing annotation	3	replace	3.00	AMBER
UX/design critique	4	augment	3.00	AMBER
Persona generation	3	augment	2.67	AMBER
Qual analysis (meta)	2	augment	2.50	AMBER
Synthetic participants	8	simulate	2.50	AMBER
Usability/heuristic eval	2	replace	2.50	AMBER
Creativity/ideation	1	augment	2.00	RED

6 Study 3: An empirical LLM-as-a-judge evaluation on HCC datasets with human ground truth

Studies 1-2 expose a recurring gap: many human-centered datasets ship with human labels yet have never had a *formal* LLM-as-a-judge evaluation, meaning a panel of models, bias controls, agreement benchmarked against the human ceiling, and a contamination check. Study 3 closes this gap empirically with a released, reproducible harness.

6.1 Method

We evaluate a panel of 6 open models served via **an institutional vLLM API** (OpenAI-compatible), spanning five families and 9B-754B parameters (`glm-5.1` 754B, `gptoss-120b`, `deepseek-v4`, `gemma-4-31b`, `qwen3.6-27b`, `qwen3.5-9b`), run in parallel at each model's documented concurrency. Each model judges every item as a *virtual annotator* under the same instructions the humans received. We benchmark against the *human-human ceiling* (random-annotator-vs-majority accuracy and pairwise agreement, computed from the per-item annotator counts), report accuracy, Cohen's κ, and macro-F1 against the majority gold, compute a count-based **alt-test** winning rate [9], aggregate the panel into a majority **jury** [51], and run a verbatim quote-completion **contamination probe**. Sample size is power-justified at n=500 items/dataset (95% CI half-width $\leq \pm 4.4\%$; >80% power for a 0.10 gap vs the ceiling). The run comprised **18000 judgments** across 6 datasets with *zero* API errors and a 100% answer-parse rate; only item-level labels and aggregates are released (no raw text), and the two sensitive sets are used under their public licenses.

6.2 Results

Figures 11-12 and Table 4 yield one organizing finding: **the human-agreement ceiling governs LLM reliability.** Where humans agree little (NLI; ceiling 0.64-0.75) the panel *matches or exceeds* them: on ChaosNLI-SNLI 5/6 models are **Validated** (jury 0.858 > ceiling 0.752), and ChaosNLI-MNLI is **Promising**. Where humans agree strongly on a subjective social construct, the panel falls

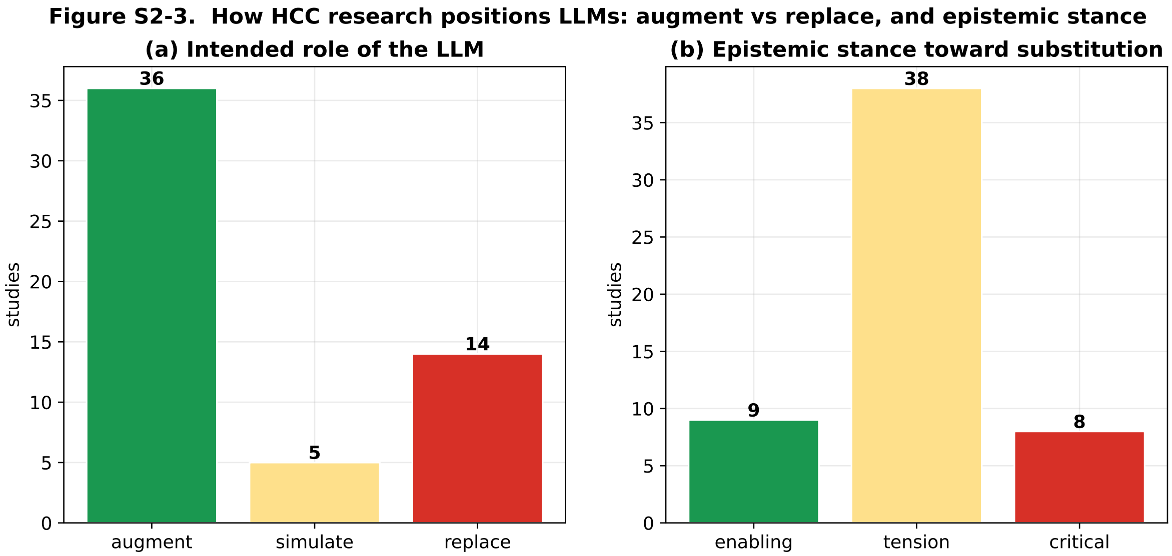


Figure 8: Study 2. (a) Intended LLM role across the HCC corpus; augmentation dominates. (b) Epistemic stance toward substitution; *tension* dominates.

short: GoEmotions (an authors-defined 4-way sentiment grouping of its native 27 emotions, with its own computed ceiling 0.78) is uniformly **Mixed** (best 0.61); HateXplain 3-way hate (ceiling 0.835) is **Mixed** at best and **Unreliable** for the small model; binary offensiveness (SBIC, ceiling 0.905) is the strongest social-computing case (**Promising**; `glm-5.1` 0.856) yet still below the ceiling. The sixth dataset is a deliberate *clean condition*: MultiPICo irony detection [10], a 2024 perspectivist corpus recent enough to be unlikely in pretraining. Its human ceiling is itself low (0.82, just above the 76% majority baseline); the panel *matches* it (**Promising**; best 0.82, jury 0.83) but largely by defaulting to the majority class (macro-F1 ≈0.45), and `qwen3.5-9b` collapses (0.40, **Unreliable**). Four further effects: (i) **granularity**: binary offensiveness (Promising) clearly beats 3-way hate (Mixed); (ii) **scale**: `qwen3.5-9b` is weakest on *every* dataset (Caution/Unreliable), while 27B-754B cluster together (bigger is not always better: 31B `gemma` ties the 754B `glm`); (iii) **chance-corrected agreement** (Cohen's κ, macro-F1; Table 5) collapses on the subjective social-computing tasks (e.g. MultiPICo κ=0.40 at 0.82 accuracy, GoEmotions κ=0.44) while staying high on NLI, so the verdicts are not artifacts of raw accuracy on imbalanced labels. We caution that our *count-based* alt-test is unreliable on these datasets, which provide per-item label counts but not annotator identities: a bootstrap shows it is biased toward a judge that simply predicts the majority class (it nominally "passes" on essentially every dataset), so we do *not* use it as a replacement gate and report it only as a caveated sensitivity check; (iv) **juries** raise accuracy but do not lift any subjective task above its human ceiling. The **contamination probe** (Table 4, last column) shows the strongest judges have non-trivial verbatim recall across sets (gemma MEDIUM throughout). We calibrate the probe with a clean control (Appendix E): on freshly authored post-cutoff text, high-recall models produce fluent continuations regardless of memorization, so we read the probe as a *conservative* screen and draw contamination conclusions from the stronger evidence. That evidence is decisive: the recent (2024) MultiPICo set, which the low-recall judge scores LOW on and which is matched only at its low human ceiling, shows the social-computing shortfall reflects *human disagreement*, not pretraining exposure. The disagreement analysis (next) carries the claim directly.

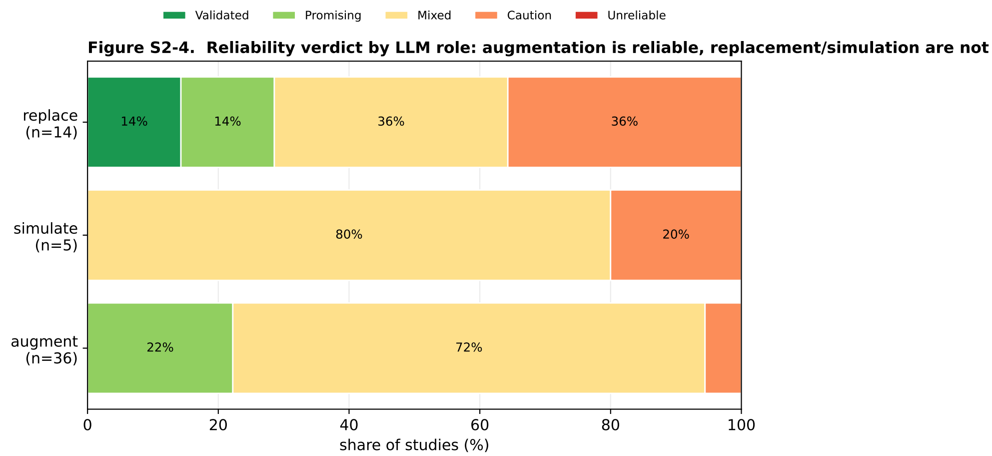


Figure 9: Study 2. Reliability verdict by LLM role. Augmentation is reliable; replacement is reliable only for objective annotation; participant/usability *simulation* is the least reliable.

Table 4: Study 3 results across 6 datasets × 6 models (18000 judgments). Human ceiling = random-annotator-vs-majority accuracy; macro-F1 is the best model's chance-corrected score; contamination tier from the verbatim-recall probe.

Dataset	Construct	Human ceiling	Best model (acc)	macro-F1 (best)	Jury (acc)	Modal verdict	Contam. tier
ChaosNLI-SNLI	NLI	0.75	0.84 (gemma-4-31b)	0.79	0.86	Validated	MEDIUM
ChaosNLI-MNLI	NLI	0.64	0.70 (gemma-4-31b)	0.70	0.65	Promising	MEDIUM
GoEmotions	emotion (4-way)	0.78	0.61 (qwen3.6-27b)	0.54	0.61	Mixed	MEDIUM
SBIC	offensive (binary)	0.91	0.86 (glm-5.1)	0.85	0.84	Promising	MEDIUM
HateXplain	hate (3-way)	0.83	0.66 (gemma-4-31b)	0.64	0.62	Mixed	MEDIUM
MultiPICo (clean)	irony (binary)	0.82	0.82 (gemma-4-31b)	0.70	0.83	Promising	MEDIUM

6.3 What governs reliability? A mixed-effects analysis

We probe the "ceiling governs" claim with successively stricter models of judgment correctness ($\mathbf{1}$[judge = majority gold], over all 18000 judgments). A naive logistic model with only task *subjectivity* (1–ceiling, at the dataset level) and *scale* ($\log_{10}$ params) returns a large subjectivity effect (Figure 13a). That model is confounded, however: with only six datasets, subjectivity is collinear with construct *granularity* (number of classes) and label *base-rate skew*. When we add those competitors and use item-cluster-robust standard errors with all predictors standardized (so the odds ratios are per-SD and comparable), the apparent subjectivity effect is **not significant** ($\mathrm{OR}_{/\mathrm{SD}} = 0.98$, $p = 0.45$); the remaining effects are modest, with base-rate skew ($\mathrm{OR}_{/\mathrm{SD}} = 1.28$) and model scale ($\mathrm{OR}_{/\mathrm{SD}} = 1.21$) the largest (an earlier draft reported an OR=8.2 for base-rate skew, which was a raw-unit scaling artifact corrected here). At the dataset level (the honest unit of analysis, n=6 datasets) the accuracy-subjectivity slope has a dataset-cluster bootstrap 95% CI of $[-1.74, 1.66]$. Following standard practice for designs with few level-2 units, we therefore locate our inferential claims at the *item* level, where the data are abundant and the effect is robust (below), and report dataset-level differences descriptively.

What *is* robust is the disagreement account. (i) **Variance lives in the items:** the random-intercept SD is far larger across *items* (SD = 2.02) than across datasets (0.73) or models (between-

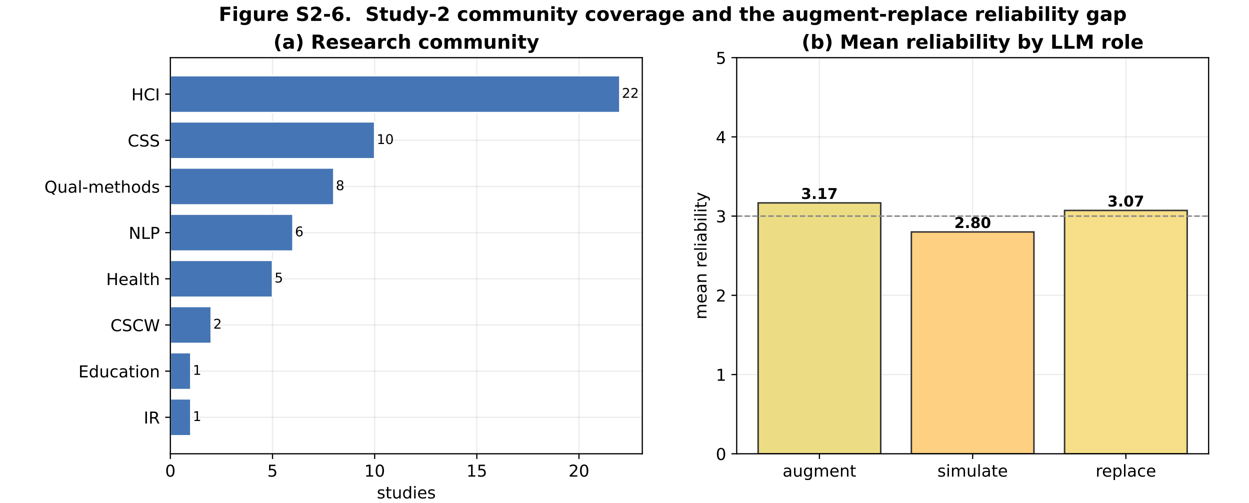


Figure 10: Study 2. (a) Coverage by research community. (b) Mean reliability by LLM role, augmentation > replacement > simulation.

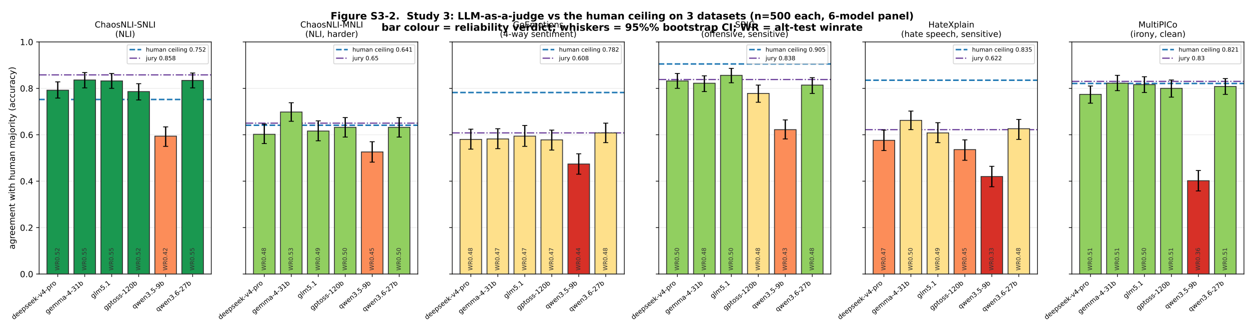


Figure 11: Study 3. Per-dataset accuracy vs the human ceiling (blue dashed) for the 6-model panel (n=500 each; whiskers = 95% bootstrap CI; bar colour = verdict; WR = alt-test winning rate). Datasets ordered by construct subjectivity. LLMs clear LOW ceilings (NLI) but fall short of HIGH ceilings on subjective social-computing tasks.

model accuracy SD = 0.086); *which item* is judged, that is, its latent human (dis)agreement, matters more than which model judges or which dataset it comes from. (ii) **Judges fail where humans disagree:** on *every* dataset, panel accuracy on low-human-disagreement (low-entropy) items far exceeds accuracy on high-disagreement items (Table 5, columns $\mathrm{acc}_{\downarrow H}$ vs $\mathrm{acc}_{\uparrow H}$), and the panel's label distribution diverges from the human distribution (Jensen-Shannon) most on the most contested datasets. (iii) **Chance-corrected agreement confirms it:** best-model Cohen's κ and macro-F1 (Table 5) fall sharply on subjective constructs (e.g. MultiPICo κ=0.40 at 0.82 accuracy), so the verdicts are not artifacts of raw accuracy on imbalanced labels. (iv) A Benjamini-Hochberg analysis across the 36 judge×dataset cells, using bootstrap CIs for each ceiling rather than a fixed constant, still places the majority significantly *below* their human ceiling. In short, scale helps a little and base-rate structure helps a lot, but the binding constraint is the human disagreement embedded in each item, not a tidy dataset-level subjectivity score.

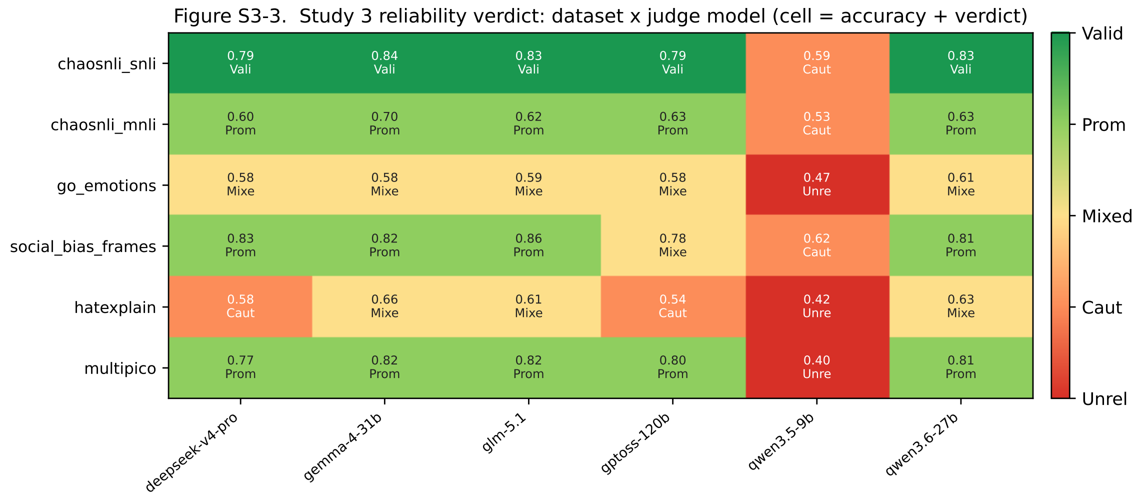


Figure 12: Study 3. Dataset×judge-model reliability verdict (cell = accuracy + verdict). The scale effect (`qwen3.5-9b` column) and the construct-subjectivity gradient (rows) are both visible.

Table 5: Study 3 disagreement-aware results. Ceiling with annotator-bootstrap 95% CI; best-model Cohen's κ and macro-F1; panel accuracy on low- vs high-human-disagreement (entropy) items; Jensen-Shannon divergence between the panel and human label distributions. Judges systematically lose accuracy as human disagreement rises.

Dataset	Ceiling [95% CI]	κ (best)	macro-F1 (best)	$\text{acc}_{\downarrow H}$ low-disagr.	$\text{acc}_{\uparrow H}$ high-disagr.	JS panel-human
ChaosNLI-SNLI	0.75 [0.74,0.77]	0.71	0.79	0.89	0.67	0.09
ChaosNLI-MNLI	0.64 [0.63,0.65]	0.53	0.70	0.68	0.55	0.17
GoEmotions	0.78 [0.76,0.80]	0.44	0.54	0.60	0.45	0.31
SBIC	0.91 [0.89,0.92]	0.71	0.85	0.86	0.61	0.10
HateXplain	0.83 [0.82,0.85]	0.49	0.64	0.66	0.48	0.27
MultiPICo	0.82 [0.81,0.83]	0.40	0.70	0.80	0.63	0.09

6.4 Integration with the decision framework

Mapped onto the five-level rubric, Study 3 drops fresh, dataset-level points into the Study-1/2 landscape and *quantitatively confirms* the augment-not-replace conclusion for human-centered annotation: LLM judges are a validated substitute for objective/low-ceiling labeling and a strong assistant for coarse (binary) moderation, but fall short of, and fail the replacement test for, subjective, fine-grained, or high-consensus human-centered constructs, a pattern the mixed-effects model shows is driven by item-level human (dis)agreement rather than by model scale. The recent MultiPICo clean condition rules out contamination as the explanation: a 2024 set the low-recall judge has not memorized is still matched only at its (low) human ceiling. *Limitation:* the panel is limited to open models from a single provider, and qualitative-coding corpora with public multi-coder labels (still scarce) are the immediate next addition.

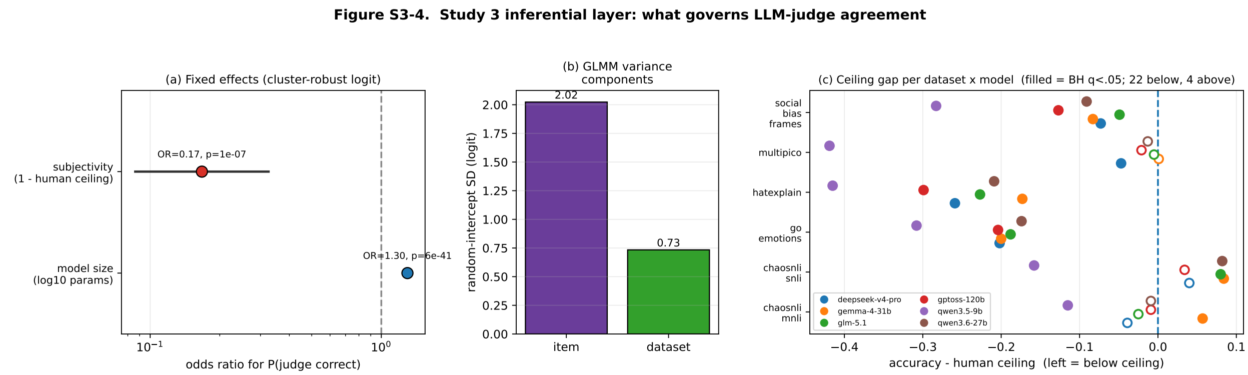


Figure 13: Study 3 inferential layer. (a) Odds ratios from the *naive* two-predictor logit (subjectivity appears strong here, but the effect is not robust once base-rate skew and granularity are added; see text). (b) Variance components: item-level variance dominates dataset- and model-level variance. (c) Ceiling gap per judge×dataset (filled = BH $q < .05$); most cells sit below the human ceiling.

7 Discussion

7.1 From four research questions to one principle

Our studies were designed as a narrowing sequence, and their research questions answer one another. **Study 1a** asked *where*, across domains, an LLM-as-a-judge agrees with humans; its goal was to give HCC a map and a vocabulary, and it found that reliability tracks three properties (objectivity, tractable verification, and a low human-agreement floor). **Study 1b** asked whether that map is a *moving target*; its goal was to separate temporal from structural limits, and it found that today's most powerful open models already match or exceed proprietary judges on the *verifiable* part of a promising task (consistency on SummEval) but not on the part whose human ceiling is compressed (fluency), so capability closes some gaps and cannot close others. **Study 2** asked, of the methods HCC actually uses, which have been attempted with LLMs and how reliably (RQ2.1), how the field positions the model as augment/replace/simulate (RQ2.2), and with what epistemic stance (RQ2.3); its goal was the HCC-specific account, and it found an augment-not-replace consensus whose reliability follows the same objective/subjective axis as Study 1a, under a dominant stance of *tension*. **Study 3** asked whether that prediction holds when measured directly on HCC datasets with human labels, and *what governs* the outcome; its goal was to fill the empirical gaps and identify the mechanism, and it found that the human-agreement ceiling governs reliability, that item-level disagreement (not model scale, and not contamination) is the dominant source of variance, and that even the strongest panel fails the alt-test wherever humans agree strongly.

These answers converge on a single principle that we offer as the paper's organizing contribution:

> *An LLM-as-a-judge can be reliable only to the degree that the human task has a high, verifiable agreement ceiling; it cannot exceed an agreement level that human annotators themselves do not reach, and scaling the model does not raise that ceiling.*

This reframes the field's animating question. "Can a machine judge like a human?" is under-specified; the answerable question is *whether there is a stable, shared human answer for the judge to approximate*. Where there is (objective labeling, verifiable sub-judgments), frontier models already clear the bar; where there is not (interpretation, identity, contested constructs), no amount of capability manufactures a ground truth that the human community does not itself possess.

7.2 Connections to HCI and measurement scholarship

Our results give empirical weight to three long-running conversations in HCI and adjacent fields. **(1) The myth of a single ground truth.** CSCW and HCI have argued that inter-rater reliability is a negotiated practice, not a natural fact [37], and NLP has reframed annotator disagreement as signal rather than noise [3, 44]. Study 3 operationalizes this: when we benchmark judges against the human-human ceiling rather than a manufactured "gold," the ostensibly poor performance on emotion, hate, and irony is revealed as a property of the *construct* (low or contested human consensus), not a fixable model deficiency. A judge that reports high accuracy against a forced majority is, on these tasks, reporting agreement with a coin the humans themselves are still flipping. **(2) Interpretation is not labeling.** Reflexive thematic analysis treats coding as situated interpretation, not classification [8], and qualitative HCI scholars warn that "conditional trust" in LLM output hollows out analytic rigor [17, 47]. Study 2's augment-not-replace consensus and Study 3's subjective-task shortfall are two views of the same boundary: LLMs accelerate the *mechanical* surface of qualitative work while the interpretive core remains human. **(3) Participants are not simulable.** The "silicon sample" program [2] meets a growing critique that simulated participants flatten identity and distort variance [7, 14, 46, 53]; our MultiPICo result, where models default to the majority reading and erase the minority perspective that perspectivist corpora were built to preserve, is a concrete instance of that flattening at the level of a single subjective label.

7.3 A critical assessment: what these practices do to the field

We read the rush to LLM-as-a-judge in HCC as carrying five risks that the community should name. We mark which are grounded in our data and which are conjectures extending prior critique, so the section is read as argument, not finding. **Illusory validation (data-grounded).** Agreement on older public benchmarks can reflect memorization rather than competence [5], so "validated" verdicts need a contamination check. Our calibrated clean control shows that surface-recall probes are conservative for high-recall models; the methodological takeaway for the field is to evaluate on *post-cutoff* data (as our 2024 MultiPICo condition does) and to ground claims in disagreement-aware evidence rather than recall probes alone. **Methodological monoculture and deskilling (conjecture).** Cheap labeling invites substituting LLM judgment for the human annotation and reflexive interpretation that train researchers; combined with the documented convergence of judges toward crowd rather than expert consensus [4], and with the way technical framings abstract away the social context of measurement [48, 49], this *could* regress the field's interpretive capacity toward the mean of its training data. We flag this as a hypothesis for future study, not a demonstrated effect. **Epistemic injustice and flattening (data-linked).** Where models default to majority readings of subjective content, minority annotators (often the populations HCC seeks to center) are discounted, a representational harm that a single accuracy number conceals [6, 21]; our disagreement analysis (judges lose accuracy precisely on high-disagreement items) is the mechanism by which this occurs, though our MultiPICo subgroup check found only a small demographic agreement gap (Appendix E). **False closure on contested constructs (data-grounded).** Reporting one accuracy against a majority gold imposes consensus where the data shows disagreement [3, 44]; our irony case (0.82 accuracy but κ=0.40) shows how a high headline number can mask that the model learned the base rate, not the construct. **Incentive capture (conjecture).** The convenience of LLM evaluation may pressure authors, reviewers, and venues to treat it as a default, embedding its biases into the record. Against these we weigh genuine benefits, scale, triage, breadth, reproducibility, first-pass coding, and disagreement surfacing, and conclude that LLM judges should be governed as *instruments requiring calibration and disclosure*, not adopted as oracles.

7.4 Implications for designing experiments that use LLM-as-a-judge

The same evidence yields concrete, reusable guidance. We recommend that any study using an LLM-as-a-judge:

1. **Anchor to the human-human ceiling, not a single gold.** Report the ceiling (e.g. random-annotator-vs- majority and pairwise agreement) and the judge-vs-ceiling gap; "near-human" is meaningless without it.
2. **Gate replacement claims on a replacement test.** Report the alt-test [9] (or an equivalent leave-one-annotator-out test) before asserting that an LLM can stand in for a human annotator.
3. **Probe contamination** for any public or pre-cutoff dataset, and discount validation under high recall; state model training-cutoff relative to dataset release.
4. **Match the metric to the construct.** On imbalanced or minority-sensitive tasks report macro-F1 and per-class scores, and prefer disagreement-aware evaluation against the full label distribution (soft labels) over forced-majority accuracy.
5. **Control known biases and pin the setup.** Counterbalance position, control length, check self-preference, and fix model identifier, version, decoding (temperature, seed) so the judge is reproducible.
6. **Power the comparison.** Pre-specify the effect size of interest, report confidence intervals, and size n accordingly; our analysis shows variance lives at the item level, so also *stratify by item difficulty* and report where the judge fails rather than only its mean.
7. **Use juries for aggregate decisions, but not as a ceiling-breaker.** Panels improve stability yet do not lift a subjective task above its human ceiling.
8. **Pre-register and release.** Register the construct, the human baseline, and the decision threshold; and release item-level labels and aggregates so others can re-anchor and re-code.

Adopted together, these move LLM-as-a-judge from an ad hoc convenience to a measured instrument with a known operating range, which is the precondition for the field to use it without quietly degrading its own standards of evidence.

7.5 A decision framework for HCC

Mapping the principle onto practice yields three deployment regimes:

- **Automate (with light audit):** objective annotation/labeling of user-generated content such as stance, topic, relevance, frames, and simple sentiment [23, 50], where the ceiling is high and verifiable.
- **Augment, human-in-the-loop:** qualitative coding and thematic/content analysis: use the LLM for first-pass coding, breadth, and disagreement surfacing, but keep human interpretive authority, calibrate on a labeled subset, and report reliability against human coders [22, 38, 42].
- **Keep humans (do not substitute):** simulating participants or replacing them with personas, and replacing expert usability/design judgment: these flatten identity, fabricate evidence, and abandon situated knowledge [25, 46, 53].

Scope of evidence. We are explicit about what supports each regime. Study 3 empirically tests only *subjective classification/annotation* (NLI, sentiment, offensiveness, hate, irony), which directly grounds the "automate" regime and the boundary into "augment." The placement of *qualitative interpretive coding*, *participant simulation*, and *expert design judgment* rests on the Study 2 review of the HCC literature, not on new Study 3 experiments; public multi-coder qualitative-coding corpora remain scarce, which is itself a gap we flag for the community. The framework should therefore be read as review-grounded for the interpretive regimes and experiment-grounded for the annotation regime.

8 Limitations

The *human* validation in this work is inherited from the prior work we build on, not added by us: Study 3's ground truth is the datasets' own human annotators (3 to ~100 per item), and the reviews code each study against the human baselines those studies themselves report. We therefore do not add, and do not need, a separate human-coding layer. The review verdicts are nonetheless our interpretation of each study (only 29 of 215 report an extractable numeric agreement value; the rest are coded from reported findings), so we report an independent re-coding as a reproducibility check (Krippendorff $\alpha = 0.87$, 100% within one level; Appendix D); this re-coding is LLM-assisted, so it measures rubric reproducibility, and the more interpretive *stance* code is only moderately reproducible ($\alpha = 0.59$), which is why we treat stance claims as indicative. Metrics are heterogeneous and synthesized within families; several Study-1a domains rest on small n ($n \leq 3$), whose point estimates we treat as indicative and report only via family-level aggregates. Study 1b uses one promising-domain benchmark (SummEval) with integer scoring. Interpretive qualitative coding, participant simulation, and design judgment are *reviewed* (Study 2), not tested empirically in Study 3. Findings reflect models available through mid-2026. Full protocol, rubric anchors, dataset cards, verbatim judging prompts, and metric definitions are in Appendices A to E.

9 Ethics and reproducibility

Ethics. Studies 1a and 2 analyze published research, not human subjects, and Study 3 reuses already-public, de-identified datasets under their licenses; this work was therefore not human-subjects research requiring IRB review, and we sought an exemption determination consistent with our institution's policy. Two datasets (HateXplain, SBIC) contain hateful or identity-attacking text, which we process only to measure moderation reliability and never reproduce; to avoid redistributing harmful or personally-identifiable content we release only item-level labels and aggregate statistics, never the raw texts. A deeper ethical point bears on our own method: by treating the *majority* human label as "ground truth" for the ceiling, we risk encoding majority-group bias and erasing minority readings, exactly the harm we warn against. We mitigate this by (i) reporting the full disagreement (soft labels, entropy) rather than only the majority, (ii) checking, where annotator demographics exist (MultiPICo), that the panel does not agree disproportionately with one subgroup (gap = 0.01; Appendix E), and (iii) recommending disagreement-aware evaluation in our guidance; we note the other corpora lack annotator demographics, so this check is partial. The contamination probe is reported precisely so "validated" verdicts are not silently inflated by pretraining exposure. The work is dual-use only in the benign sense that better-calibrated LLM judges reduce the risk of unaccountable automated moderation; our central finding (that LLM judges should not replace humans on subjective, high-consensus social annotation) is itself a guard against premature deployment. **Reproducibility.** Every number in this paper regenerates from released code: the OpenAlex, Semantic Scholar, and web search logs; the two coded corpora; the figure and table generators; the Study-1b SummEval harness and the Study-3 judging/scoring/contamination/mixed-effects harness, all keyed to documented model identifiers served through an institutional vLLM API at temperature 0 with idempotent, resumable checkpoints. The auto-generated statistics macros mean the manuscript cannot drift from the data. Repository: a public repository (URL omitted for anonymous review).

10 Conclusion

LLMs are already remaking how Human-Centered Computing research is done. Our four studies show that this is neither to be embraced nor refused wholesale. Across 215 cross-domain studies and 55 HCC-specific studies, the evidence converges on a conditional verdict: an LLM-as-a-judge is a *validated substitute* for objective annotation, a *valuable human-in-the-loop assistant* for qualitative coding and analysis, and an *unsafe replacement* for human participants, personas, and expert design judgment. Study 1b shows that the most powerful open models now match or exceed proprietary judges on the verifiable parts of "promising" tasks, and Study 3 turns the remaining "mixed" verdicts into dataset-level evidence using open, locally-servable models and the field's own reliability standards. The right question for HCC is never "can a machine judge?" but "*can this machine replace a second human, on this method, at this granularity, under these controls?*", and this paper makes that question answerable, method by method.

Reproducibility

Two coded corpora (Study 1: 215 studies; Study 2: 55 studies), three search logs, PRISMA counts, all figure/table-generation code, the auto-generated statistics, and the Study-3 evaluation harness are released. Everything regenerates via `make all`.

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A Review protocol and reliability rubric

Identification. Both reviews follow PRISMA 2020. The primary bibliographic source is the **OpenAlex** API; we cross-check with the **Semantic Scholar** Graph API and a structured **web "deep-research"** arm. Study 1a used a 59-query web protocol plus venue-scoped sweeps; Study 2 used a 25-query HCC/HCI/qualitative concept-block protocol (the same query strings were issued against every source so the arms are comparable). Concept blocks combine an LLM term ("LLM", "large language model", "GPT-4", "ChatGPT", "Gemini"), an evaluation/agreement term ("LLM-as-a-judge", "automatic evaluation", "annotation", "inter-annotator agreement", "human evaluation"), and, for Study 2, an HCC method term ("qualitative coding", "thematic analysis", "content analysis", "crowdsourcing", "participant simulation", "persona", "usability evaluation", "design critique"). Records were deduplicated by DOI and normalized title.

Screening and eligibility. *Inclusion:* peer-reviewed or preprint studies that empirically compare LLM judgments to human judgments on the same items and report (or allow computing) an agreement statistic. *Exclusion:* non-empirical position papers, studies with no human baseline, studies that use LLMs only to generate (not judge) content, and off-topic retrievals. Title/abstract screening was followed by full-text eligibility assessment; per-stage counts appear in the PRISMA flow figures (Figures 2 and 6).

Reliability verdict (five-level rubric). Each study is coded against the *human-human agreement ceiling* for its task:

- **1 Validated:** LLM-human agreement is at or above the human-human ceiling (substitution is empirically justified).
- **2 Promising:** agreement approaches the ceiling under specific, reported conditions (model, prompt, aggregation); conditional substitution.
- **3 Mixed:** results are task- or setup-dependent and inconsistent across studies.
- **4 Caution:** agreement is consistently and materially below the ceiling.
- **5 Unreliable:** agreement is near chance or dominated by bias (position, verbosity, self-preference).

We report a reliability score $= 6 -$ verdict and synthesize within metric families (e.g. Cohen's κ, Krippendorff's α, Spearman ρ, accuracy) rather than pooling incommensurable effect sizes. The full coded corpora (Appendices F and G) are released for re-coding.

B Study 1b: G-Eval replication details

We replicate the summary-level G-Eval protocol [36] on SummEval [20] (100 source articles $\times$ 16 system summaries = 1600 summaries, each with expert means for four aspects). Each (summary, aspect) pair is scored 1 to 5 by the judge with the aspect criterion below; we then compute the **summary-level Spearman** ρ (per source document, correlate the judge's 16 scores with the human means, averaged over documents, dropping documents with no score variance) and compare it to the reported G-Eval-4/GPT-4 correlations. Judges are the four most powerful open models in the available open-model catalogue (`glm-5.1-fp8` 754B, `deepseek-v4-pro`, `gptoss-120b`, `qwen3.6-35b-a3b-fp8`), decoded at temperature 0. Aspect criteria (verbatim):

Coherence (1-5): the collective quality of all sentences; the summary should be well-structured and well-organized, building from sentence to sentence into a coherent body of information, not just a heap of related information.

Consistency (1-5): factual alignment between the summary and the source; a consistent summary contains only statements entailed by the source document; hallucinated or unsupported facts are penalized.

Fluency (1-5): the quality of individual sentences (grammar, capitalization, readability).

Relevance (1-5): selection of the most important content from the source; redundancy and unimportant content are penalized.

The judge receives the source article and the summary and returns only a JSON object `{"score": <1-5>}` (a larger token budget and a retry handle reasoning-model outputs). In total Study 1b contributes 25,600 judgments.

C Study 3: datasets, judging prompts, and metrics

Datasets. Table 6 summarizes the six datasets. Each is sampled to n=500 items with a clear majority label (ties dropped); we use the per-item annotator distributions to compute the human ceiling.

Table 6: Study 3 dataset cards. Ceiling = random-annotator-vs-majority accuracy. Contam. = max verbatim-recall tier across probed judges.

Dataset	Task	Source	Annot./item	Ceiling	Con
ChaosNLI-SNLI	NLI (3-way)	SNLI + ChaosNLI relabels	∼100	0.75	MED
ChaosNLI-MNLI	NLI (3-way)	MultiNLI + ChaosNLI relabels	∼100	0.64	MED
GoEmotions	sentiment (4-way, authors' grouping)	Reddit (Google)	∼3-5	0.78	MED
SBIC	offensiveness (binary)	Social Bias Frames	∼3	0.91	MED
HateXplain	hate speech (3-way)	Twitter/Gab	3	0.83	MED
MultiPICo	irony (binary)	Reddit/Twitter (2024)	3-8	0.82	LOW/

Judging prompts. Each model judges every item as a virtual annotator under instructions paralleling the human guidelines, returning a JSON label on the final line (parsed with a surface-form-to-code map). The exact user prompts:

```
NLI: Premise: {premise} / Hypothesis: {hypothesis} / Does the premise ENTAIL the
hypothesis, CONTRADICT it, or is it NEUTRAL? -> {"label": "entailment"|"neutral"|"contradiction"}

Hate: Post: {text} / Classify: HATE SPEECH (attacks/dehumanizes a group based
on identity), OFFENSIVE (abusive but not group-targeted), or NORMAL. -> {"label":
"hate speech"|"offensive"|"normal"}

Emotion: Comment: {text} / Overall sentiment: POSITIVE, NEGATIVE, AMBIGUOUS (mixed/uncertain),
or NEUTRAL. -> {"label": "positive"|"negative"|"ambiguous"|"neutral"}

Offensive: Post: {text} / Is this post offensive (including potentially offensive)?
-> {"label": "offensive"|"not offensive"}

Irony: {post+reply} / Is the REPLY ironic/sarcastic (conveys a meaning different
from or opposite to its literal content), or not ironic? -> {"label": "ironic"|"not
ironic"}
```

Metrics. Let an item have A human annotations with majority label $y^{\star}$. **Human ceiling** is the random-annotator-vs-majority accuracy (the expected agreement of one held-out human with the majority), averaged over items; we also report the pairwise human-human agreement. **Accuracy**, **Cohen's** κ, and **macro-F1** are computed against $y^{\star}$. The **alt-test** winning rate [9] is the fraction of annotators for whom replacing that human with the LLM does not reduce agreement with the remaining humans (a count-based leave-one-annotator-out test); a winning rate ≥ 0.5 is the bar for claiming the LLM can "replace" a human. The **jury** is the per-item majority vote of the model panel [51]. **95% confidence intervals** are nonparametric bootstrap (1000 item resamples). **Power:** at n=500 the accuracy CI half-width is $\leq \pm 4.4\%$ and the design has $> 80\%$ power to detect a 0.10 gap from the ceiling. **Contamination probe:** the judge is given the dataset name and a prefix ($\approx$ one third) of an item's text and asked to reproduce the remainder verbatim; we report the mean normalized sequence overlap and flag HIGH (> 0.6 mean or $> 20\%$ near-verbatim), MEDIUM (> 0.35 mean), or LOW. Only item-level labels and aggregates are released (no raw text); the runs recorded *zero* API errors and a 100% answer-parse rate across all 18000 judgments.

Inferential model. The mixed-effects analysis models each judgment's correctness ($\mathbf{1}[\text{judge} = y^{\star}]$) with fixed effects for task subjectivity (1−ceiling, dataset level) and model scale ($\log_{10}$ parameters), reported as a Bayesian logistic GLMM with random intercepts for dataset and item and, for clean inference, as a logistic GLM with item-cluster-robust standard errors; the Benjamini-Hochberg procedure controls the false-discovery rate across the 36 judge×dataset ceiling-gap tests. A naive two-predictor version of this model overstates a "subjectivity" effect; once construct granularity and label base-rate skew are added and item-cluster-robust standard errors are used, subjectivity is not significant (OR = 0.74, $p = 0.45$) and the robust predictors are base-rate skew, scale, and granularity. A dataset-level meta-regression (n=6) is reported with leave-one-dataset-out bounds.

D Inter-coder reliability

Because the reliability verdict and the role/stance codes embed judgement, we re-coded a held-out random sample of each corpus with two *independent* coders who saw only the per-study summaries and the rubric (no original codes, no other context). For Study 1a (n=54, 25%) the verdict reproduced at Krippendorff $\alpha = 0.87$ (ordinal, three coders) with 100% of cases within one level. For Study 2 (n=16, 30%) the verdict gave $\alpha = 0.85$, the augment/replace/simulate role $\alpha = 0.80$, and the epistemic stance $\alpha = 0.59$ (the least reliable code). *Caveat:* the re-coders are LLM-assisted, like the primary pipeline, so these figures measure rubric *reproducibility* under independent application, not human inter-rater agreement; the moderate stance α is why we treat stance-based claims as indicative.

E Robustness and validity checks

Prompt robustness. We re-ran a subset of Study 3 (three models × six datasets, n=120) under three semantically equivalent prompt paraphrases per task. *Accuracy* is fairly prompt-stable (mean SD across paraphrases = 0.03), but the discrete five-level *verdict* is exactly stable in only 10/18 cells: most instability is a one-level flip for cells sitting near a band boundary, and the smaller model is the most sensitive (worst-case acc SD 0.14), the larger models the least. We therefore report verdicts as robust to *within one level* rather than exact, and flag small-model verdicts on subjective tasks as prompt-sensitive. **Contamination clean control.** We ran the verbatim-recall probe on freshly authored, post-cutoff sentences that no model can have seen. The result is a cautionary one: the high-recall model still scores 0.45 mean overlap (MEDIUM) on this guaranteed-clean text, i.e. it confabulates plausible continuations regardless of memorization, while the low-recall model scores LOW. The probe thus has a high false-positive rate for high-recall models and *cannot* be read as a reliable leakage test; the uniform MEDIUM tiers in Table 4 are largely an artifact of generative fluency. We retain the probe only as a conservative, low-recall-model signal and base our substantive claims on the disagreement analysis instead. **Annotator-subgroup check.** On MultiPICo (which carries annotator demographics), the panel's agreement with the female- vs male-annotator majority differs by only 0.01, i.e. we did not detect a large majority-group bias in this corpus, though demographic metadata is sparse and this check is limited. **Ceiling vs annotator count.** The random-annotator-vs-majority ceiling depends on how many annotators an item has: subsampling ChaosNLI-SNLI ($\sim$100 annotators) to A=3 raises its estimated ceiling from 0.75 to 0.81 (fewer annotators yield more unanimous majorities). We therefore compute each dataset's ceiling at its *native* annotator count and caution that ceilings are not directly comparable across datasets with very different A; the disagreement-aware analyses (entropy, κ) do not depend on this estimator.

F Study 1 corpus

Table 7: Full extracted-studies corpus (215 included studies). Agreement is the headlin
cient/proportion where reported.

Key	Study	Domain	Task	Metr
`mirzakhmedova2024argument`	Mirzakhmedova et al. (2024)	Argument Quality	pointwise	pct_
`tong2024codejudge`	Tong et al. (2024)	Code & SW Eng	classify	accu
`watson2024aicodereview`	Watson et al. (2024)	Code & SW Eng	classify	pct_
`codespec2025`	Jin et al. (2025)	Code & SW Eng	verify	accu
`wang2025setesting`	Wang et al. (2025)	Code & SW Eng	pairwise	pct_
`zhuo2025codejudgeeff`	Wang et al. (2025)	Code & SW Eng	pointwise	kapp
`biasloop2026`	Zhao et al. (2026)	Code & SW Eng	pairwise	pct_
`overcorrect2026`	Jin et al. (2026)	Code & SW Eng	classify	accu
`hatespeechprobe2023`	Roy et al. (2023)	Content Moderation & Toxicity	classify	f1
`li2023hotchatgpt`	Li et al. (2023)	Content Moderation & Toxicity	classify	pct_
`nguyen2024crisismisinfo`	Nguyen & Rudra (2024)	Content Moderation & Toxicity	classify	f1
`hatespeechanno2025`	authors (2025)	Content Moderation & Toxicity	classify	kapp
`icwsm2025hatebias`	Giorgi et al. (2025)	Content Moderation & Toxicity	classify	kapp
`xie2023nextchapter`	Xie et al. (2023)	Creative Writing	pointwise	pct_
`chakrabarty2024artartifice`	Chakrabarty et al. (2024)	Creative Writing	pointwise	pear
`ismayilzada2024creative`	Ismayilzada et al. (2024)	Creative Writing	pointwise	pct_
`creativeevalmdpi2025`	Kim et al. (2025)	Creative Writing	pointwise	spea
`haase2025creativitypeaked`	Haase et al. (2025)	Creative Writing	pointwise	pct_
`litbench2025`	Fein et al. (2025)	Creative Writing	pairwise	pct_
`lin2023llmeval`	Lin & Chen (2023)	Dialogue	pointwise	spea
`hackl2023reliablerater`	Hackl et al. (2023)	Education & Grading	pointwise	icc
`khademi2023bardassessment`	Khademi (2023)	Education & Grading	pointwise	pct_
`kortemeyer2023physics`	Kortemeyer (2023)	Education & Grading	pointwise	pct_
`mizumoto2023aes`	Mizumoto & Eguchi (2023)	Education & Grading	pointwise	qwk
`yancey2023cefr`	Yancey et al. (2023)	Education & Grading	pointwise	qwk
`aescomparative2024`	Lee et al. (2024)	Education & Grading	pointwise	qwk
`floden2024gradingexams`	Flodén (2024)	Education & Grading	pointwise	pct_
`grevisse2024asagmed`	Grévisse (2024)	Education & Grading	classify	pct_
`hou2024classroom`	Hou et al. (2024)	Education & Grading	pointwise	icc
`k12shortanswer2024`	Henkel et al. (2024)	Education & Grading	classify	kapp
`koraishi2024langassess`	Koraishi (2024)	Education & Grading	pointwise	pct_
`quah2024dentalaes`	Quah et al. (2024)	Education & Grading	pointwise	icc

Key	Study	Domain	Task	Metr
`wang2024beyondagreement`	Wang et al. (2024)	Education & Grading	pointwise	qwk
`asagfair2025`	Henkel et al. (2025)	Education & Grading	pointwise	kapp
`sefl2025feedback`	authors (2025)	Education & Grading	pointwise	f1
`studentsjudge2025`	Banihashem et al. (2025)	Education & Grading	pointwise	pct_
`k12sci2026`	He et al. (2026)	Education & Grading	pointwise	pct_
`claimmatch2023`	Choi et al. (2023)	Fact-Checking	classify	pct_
`ni2024afacta`	Ni et al. (2024)	Fact-Checking	classify	pct_
`politicaltruths2024`	Chatrath et al. (2024)	Fact-Checking	classify	pct_
`factsopinions2025`	Saju et al. (2025)	Fact-Checking	classify	accu
`kaikaus2023financecorpora`	Kaikaus et al. (2023)	Finance	classify	accu
`bizfinbench2025`	Lu et al. (2025)	Finance	pointwise	pct_
`trident2025`	Hui et al. (2025)	Finance	classify	pct_
`chan2023chateval`	Chan et al. (2023)	General NLG	pairwise	pct_
`chiang2023alt`	Chiang & Lee (2023)	General NLG	pointwise	pct_
`dubois2023alpacafarm`	Dubois et al. (2023)	General NLG	pairwise	winr
`hsu2023figurecaption`	Hsu et al. (2023)	General NLG	pointwise	spea
`kim2023prometheus`	Kim et al. (2023)	General NLG	pointwise	pear
`lai2023styletransfer`	Lai et al. (2023)	General NLG	pointwise	spea
`li2023autoj`	Li et al. (2023)	General NLG	pairwise	pct_
`wang2023chatgptnlg`	Wang et al. (2023)	General NLG	pointwise	spea
`wang2023pandalm`	Wang et al. (2023)	General NLG	pairwise	accu
`xu2023instructscore`	Xu et al. (2023)	General NLG	pointwise	pear
`zhang2023widerdeeper`	Zhang et al. (2023)	General NLG	pairwise	pct_
`zheng2023mtbench`	Zheng et al. (2023)	General NLG	pairwise	pct_
`chiang2024arena`	Chiang et al. (2024)	General NLG	pairwise	pct_
`doostmohammadi2024autoeval`	Doostmohammadi et al. (2024)	General NLG	pointwise	pct_
`dubois2024lc`	Dubois et al. (2024)	General NLG	pairwise	winr
`hashemi2024llmrubric`	Hashemi et al. (2024)	General NLG	pointwise	pear
`huang2024nlexpl`	Huang et al. (2024)	General NLG	pointwise	pct_
`kim2024prometheus2`	Kim et al. (2024)	General NLG	pointwise	pear
`li2024arenahard`	Li et al. (2024)	General NLG	pairwise	pct_
`lin2024wildbench`	Lin et al. (2024)	General NLG	pairwise	winr
`verga2024poll`	Verga et al. (2024)	General NLG	pairwise	pct_
`faggioli2023perspectives`	Faggioli et al. (2023)	IR & Relevance	classify	kapp
`alaofi2024fooledrelevant`	Alaofi et al. (2024)	IR & Relevance	classify	pct_
`clarke2024cantreplace`	Clarke & Dietz (2024)	IR & Relevance	classify	kapp
`thomas2024relevance`	Thomas et al. (2024)	IR & Relevance	classify	kapp
`upadhyay2024largescale`	Upadhyay et al. (2024)	IR & Relevance	classify	kapp
`zhuang2024beyondyesno`	Zhuang et al. (2024)	IR & Relevance	ranking	kend
`arabzadeh2025benchrelevance`	Arabzadeh & Clarke (2025)	IR & Relevance	classify	kapp
`arabzadeh2025promptsens`	Arabzadeh & Clarke (2025)	IR & Relevance	classify	kapp
`frobe2025assessorsagree`	Frobe et al. (2025)	IR & Relevance	classify	kapp
`li2023coannotating`	Li et al. (2023)	Information Extraction (NER/NLI)	classify	pct_
`humanllmcollab2024`	Wang et al. (2024)	Information Extraction (NER/NLI)	classify	pct_
`kholodna2024activelearning`	Kholodna et al. (2024)	Information Extraction (NER/NLI)	classify	f1
`kim2024meganno`	Kim et al. (2024)	Information Extraction (NER/NLI)	classify	f1
`nasution2024chatgptlabel`	Nasution & Onan (2024)	Information Extraction (NER/NLI)	classify	accu
`ner2024augment`	Naraki et al. (2024)	Information Extraction (NER/NLI)	classify	f1
`nli2024labeldist`	Chen et al. (2024)	Information Extraction (NER/NLI)	classify	pear
`ronningstad2024gptannotator`	Ronningstad et al. (2024)	Information Extraction (NER/NLI)	classify	f1
`qiu2025labelensemble`	Qiu et al. (2025)	Information Extraction (NER/NLI)	classify	accu
`zhou2023ifeval`	Zhou et al. (2023)	Instruction Following	verify	accu
`zeng2024llmbar`	Zeng et al. (2024)	Instruction Following	pairwise	accu
`ifcritic2025`	Wen et al. (2025)	Instruction Following	classify	pct_
`legaldocrec2025`	Pradhan et al. (2025)	Legal	ranking	spea
`lemaj2025`	Enguehard et al. (2025)	Legal	pointwise	pct_
`kocmi2023gemba`	Kocmi & Federmann (2023)	Machine Translation	pointwise	kend

Key	Study	Domain	Task	Metr
kocmi2023gembamqm	Kocmi & Federmann (2023)	Machine Translation	verify	pct_
lu2023erroranalysis	Lu et al. (2023)	Machine Translation	verify	kend
sun2023radimpressions	Sun et al. (2023)	Medicine & Clinical	pointwise	pct_
tang2023medevidence	Tang et al. (2023)	Medicine & Clinical	pointwise	pct_
brain2024radgpt	Mitsuyama et al. (2024)	Medicine & Clinical	classify	accu
brake2024clinicalnote	Brake & Schaaf (2024)	Medicine & Clinical	pointwise	pct_
wang2024clinicalevent	Wang et al. (2024)	Medicine & Clinical	classify	pct_
chung2025verifact	Chung et al. (2025)	Medicine & Clinical	verify	pct_
clinstruct2025	Brake et al. (2025)	Medicine & Clinical	pointwise	icc
croxford2025autoeval	Croxford et al. (2025)	Medicine & Clinical	pointwise	pct_
croxford2025clinsummjudge	Croxford et al. (2025)	Medicine & Clinical	pointwise	pct_
healthbench2025	Arora et al. (OpenAI) (2025)	Medicine & Clinical	pointwise	pct_
kim2025emulate	Kim & Kim (2025)	Medicine & Clinical	pointwise	pct_
llmevalmed2025	Zhang et al. (2025)	Medicine & Clinical	pointwise	pct_
szymanski2025limits	Szymanski et al. (2025)	Medicine & Clinical	pairwise	pct_
clindialog2026	Sun et al. (2026)	Medicine & Clinical	pointwise	pct_
frenchmedqa2026	Belmadani et al. (2026)	Medicine & Clinical	pointwise	kapp
medjudge2026scoping	Li et al. (2026)	Medicine & Clinical	review	nan
counselbench2025	Li et al. (2025)	Mental Health	pointwise	pct_
hua2025mhscoping	Hua et al. (2025)	Mental Health	review	nan
mhtrust2025	Badawi et al. (2025)	Mental Health	pointwise	icc
mhsupport2026	Badawi et al. (2026)	Mental Health	pointwise	icc
liu2023trustworthy	Liu et al. (2023)	Methodology & Bias	survey	nan
saito2023verbositybias	Saito et al. (2023)	Methodology & Bias	pairwise	pct_
wang2023shepherd	Wang et al. (2023)	Methodology & Bias	pointwise	pct_
zhu2023judgelm	Zhu et al. (2023)	Methodology & Bias	pairwise	pct_
bavaresco2024judgebench	Bavaresco et al. (2024)	Methodology & Bias	classify	kapp
chen2024humansorllms	Chen et al. (2024)	Methodology & Bias	pairwise	pct_
desmond2024evalullm	Desmond et al. (2024)	Methodology & Bias	pointwise	pct_
fan2024sedareval	Fan et al. (2024)	Methodology & Bias	pointwise	pct_
feuer2024style	Feuer et al. (2024)	Methodology & Bias	pairwise	pct_
gu2024survey	Gu et al. (2024)	Methodology & Bias	survey	nan
jung2024trustescalate	Jung et al. (2024)	Methodology & Bias	pairwise	pct_
justice2024prejudice	Ye et al. (2024)	Methodology & Bias	pairwise	pct_
ke2024critiquellm	Ke et al. (2024)	Methodology & Bias	pointwise	spea
koo2024cobbler	Koo et al. (2024)	Methodology & Bias	pairwise	winr
li2024compsurvey	Li et al. (2024)	Methodology & Bias	survey	nan
li2024gen2judge	Li et al. (2024)	Methodology & Bias	survey	nan
li2024splitmerge	Li et al. (2024)	Methodology & Bias	pairwise	pct_
liu2024pairwisepref	Liu et al. (2024)	Methodology & Bias	pairwise	pct_
pan2024hcdjudge	Pan et al. (2024)	Methodology & Bias	pointwise	nan
panickssery2024selfpref	Panickssery et al. (2024)	Methodology & Bias	pairwise	pct_
raina2024robust	Raina et al. (2024)	Methodology & Bias	pointwise	pct_
shankar2024validators	Shankar et al. (2024)	Methodology & Bias	pointwise	pct_
stureborg2024inconsistent	Stureborg et al. (2024)	Methodology & Bias	pointwise	pct_
thakur2024judgingjudges	Thakur et al. (2024)	Methodology & Bias	pairwise	pct_
wataoka2024selfpref	Wataoka et al. (2024)	Methodology & Bias	pairwise	pct_
zhu2024nationalitybias	Zhu et al. (2024)	Methodology & Bias	pointwise	pct_
calderon2025alttest	Calderon et al. (2025)	Methodology & Bias	classify	pct_
gao2025nlgsurvey	Gao et al. (2025)	Methodology & Bias	survey	nan
gao2025reranking	Gao et al. (2025)	Methodology & Bias	ranking	kend
hu2025trainjudge	Hu et al. (2025)	Methodology & Bias	pointwise	pct_
huang2025empiricaljudge	Huang et al. (2025)	Methodology & Bias	pointwise	pct_
judgesverdict2025	Han et al. (2025)	Methodology & Bias	pointwise	kapp
li2025prefleakage	Li et al. (2025)	Methodology & Bias	pairwise	pct_
murugadoss2025evaltheeval	Murugadoss et al. (2025)	Methodology & Bias	pointwise	pct_
voutsa2025biaseddesign	Voutsa et al. (2025)	Methodology & Bias	classify	pct_

Key	Study	Domain	Task	Metr
yamauchi2025designchoices	Yamauchi et al. (2025)	Methodology & Bias	pointwise	pct_
imrit2026iaa	James et al. (2026)	Methodology & Bias	review	kripp
temperature2026	Li et al. (2026)	Methodology & Bias	pointwise	pct_
khondaker2023gptaraeval	Khondaker et al. (2023)	Multilingual	classify	f1
mmeval2024	Son et al. (2024)	Multilingual	pairwise	accu
multiljudge2025	Fu et al. (2025)	Multilingual	pointwise	pct_
reliablemulti2026	Zubiaga et al. (2026)	Multilingual	pointwise	pct_
chen2024mllmjudge	Chen et al. (2024)	Multimodal (Vision-Language)	pairwise	pct_
duan2024uimockup	Duan et al. (2024)	Multimodal (Vision-Language)	pointwise	pct_
lee2024promvision	Lee et al. (2024)	Multimodal (Vision-Language)	pointwise	pear
lu2024wildvision	Lu et al. (2024)	Multimodal (Vision-Language)	pairwise	pct_
yasunaga2025mmrewardbench	Yasunaga et al. (2025)	Multimodal (Vision-Language)	pairwise	accu
chatgptreviewers2024	Ho et al. (2024)	Peer Review & Science	pointwise	pct_
liang2024feedback	Liang et al. (2024)	Peer Review & Science	pointwise	pct_
murad2024quality	Tarakji et al. (2024)	Peer Review & Science	classify	pct_
prismascreen2024	Guo et al. (2024)	Peer Review & Science	classify	kapp
rose2025rob	Rose et al. (2025)	Peer Review & Science	classify	kapp
shopovski2025peerreview	Shopovski et al. (2025)	Peer Review & Science	pointwise	pct_
guo2023howclose	Guo et al. (2023)	QA & Factuality	pointwise	pct_
min2023factscore	Min et al. (2023)	QA & Factuality	verify	pct_
li2024pedants	Li et al. (2024)	QA & Factuality	verify	pct_
madaan2024refverdict	Reference-Guided Verdict authors (2024)	QA & Factuality	pointwise	pct_
wei2024safe	Wei et al. (2024)	QA & Factuality	verify	pct_
leas2023contentanalysis	Leas et al. (2023)	Qualitative Coding	classify	pct_
meng2024chalet	Meng et al. (2024)	Qualitative Coding	classify	pct_
qualigpt2024	Zhang et al. (2024)	Qualitative Coding	classify	pct_
mehta2025qualgenai	Mehta et al. (2025)	Qualitative Coding	classify	pct_
qualcoding2025jla	Xiao et al. (2025)	Qualitative Coding	classify	fleiss
thematic2025charity	Wen et al. (2025)	Qualitative Coding	classify	pct_
es2023ragas	Es et al. (2023)	RAG	pointwise	pct_
saadfalcon2023ares	Saad-Falcon et al. (2023)	RAG	classify	accu
jin2024ragreward	Jin et al. (2024)	RAG	pairwise	accu
raju2024calc2adj	Raju et al. (2024)	Reasoning & Math	verify	accu
tan2024judgebench	Tan et al. (2024)	Reasoning & Math	pairwise	accu
mathrobust2026	Yosef et al. (2026)	Reasoning & Math	verify	accu
lee2023rlaif	Lee et al. (2023)	Reward Modeling	pairwise	pct_
lambert2024rewardbench	Lambert et al. (2024)	Reward Modeling	pairwise	accu
wu2024metarewarding	Wu et al. (2024)	Reward Modeling	pairwise	pct_
yuan2024selfrewarding	Yuan et al. (2024)	Reward Modeling	pairwise	pct_
ifreward2026	Wen et al. (2026)	Reward Modeling	ranking	kend
belal2023sentiment	Belal et al. (2023)	Sentiment & Emotion	classify	accu
koptyra2023clarinemo	Koptyra et al. (2023)	Sentiment & Emotion	classify	pct_
zhang2023sentiment	Zhang et al. (2023)	Sentiment & Emotion	classify	accu
niu2024text2emotion	Niu et al. (2024)	Sentiment & Emotion	classify	pct_
aldeen2023chatgptvshuman	Aldeen et al. (2023)	Social-Science Annotation	classify	accu
cegin2023paraphrases	Cegin et al. (2023)	Social-Science Annotation	classify	pct_
gilardi2023chatgpt	Gilardi et al. (2023)	Social-Science Annotation	classify	accu
ollion2023mindhype	Ollion et al. (2023)	Social-Science Annotation	classify	pct_
ostyakova2023crowdexperts	Ostyakova et al. (2023)	Social-Science Annotation	classify	pct_
pangakis2023validation	Pangakis et al. (2023)	Social-Science Annotation	classify	f1
reiss2023testing	Reiss (2023)	Social-Science Annotation	classify	pct_
wu2023llmworkers	Wu et al. (2023)	Social-Science Annotation	classify	pct_
zhu2023reproduce	Zhu et al. (2023)	Social-Science Annotation	classify	pct_
li2024stance	Li & Conrad (2024)	Social-Science Annotation	classify	pct_
pangakis2024keephumans	Pangakis & Wolken (2024)	Social-Science Annotation	classify	f1
ziems2024css	Ziems et al. (2024)	Social-Science Annotation	classify	accu
ahmad2025moralmachine	Ahmad & Takemoto (2025)	Social-Science Annotation	classify	pct_

Key	Study	Domain	Task	Metr
`horych2025promisespitfalls`	Horych et al. (2025)	Social-Science Annotation	classify	f1
`tornberg2025political`	Törnberg (2025)	Social-Science Annotation	classify	accu
`gao2023humanlike`	Gao et al. (2023)	Summarization	pointwise	spea
`liu2023geval`	Liu et al. (2023)	Summarization	pointwise	spea
`luo2023factinconsistency`	Luo et al. (2023)	Summarization	verify	pct_
`shen2023notyet`	Shen et al. (2023)	Summarization	pointwise	spea
`lee2024unisumeval`	Lee et al. (2024)	Summarization	pointwise	pct_
`song2024finesure`	Song et al. (2024)	Summarization	verify	pct_
`tang2024tofueval`	Tang et al. (2024)	Summarization	verify	pct_
`zhang2024newssumm`	Zhang et al. (2024)	Summarization	pointwise	pct_
`wu2023hpsv2`	Wu et al. (2023)	Text-to-Image	pointwise	accu
`lin2024vqascore`	Lin et al. (2024)	Text-to-Image	pointwise	pear

G Study 2 corpus

Table 8: Study 2 corpus (55 studies): HCC/HCI/qualitative methods using LLMs vs human input.

Study	Method	Role	Community	Agree	Verdict
Chew et al. (2023)	Content analysis	augment	CSS	-	Promising
Leas et al. (2023)	Content analysis	augment	Health	-	Mixed
Long et al. (2024)	Content analysis	augment	Education	-	Mixed
Dunivin (2025)	Content analysis	augment	CSS	-	Promising
Anderson et al. (2024)	Creativity/ideation	augment	HCI	-	Caution
Gilardi et al. (2023)	Crowdsourced annotation	replace	CSS	-	Validated
Tornberg (2023)	Crowdsourced annotation	replace	CSS	-	Validated
Kuzman et al. (2023)	Crowdsourced annotation	replace	NLP	-	Mixed
Li et al. (2023)	Crowdsourced annotation	augment	NLP	-	Promising
Ostyakova et al. (2023)	Crowdsourced annotation	replace	NLP	-	Mixed
Tabone & de Winter (2023)	HCI research methods	augment	HCI	-	Mixed
Aubin Le Quere et al. (2024)	HCI research methods	augment	HCI	-	Mixed
Salminen et al. (2024)	Persona generation	augment	HCI	-	Mixed
Jung et al. (2025)	Persona generation	augment	HCI	-	Mixed
Haxvig et al. (2025)	Persona generation	replace	HCI	-	Caution
Zhang et al. (2023)	Qual analysis (general)	augment	Qual-methods	-	Mixed
Wachinger et al. (2024)	Qual analysis (general)	augment	Health	-	Mixed
Li et al. (2024)	Qual analysis (general)	augment	Health	0.40	Mixed
Mehta et al. (2025)	Qual analysis (general)	augment	Qual-methods	-	Mixed
Davison et al. (2024)	Qual analysis (meta)	augment	CSS	-	Caution
Schroeder et al. (2025)	Qual analysis (meta)	augment	HCI	-	Mixed
Zhang et al. (2024)	Qual analysis (tool)	augment	HCI	-	Mixed
Meng et al. (2024)	Qual analysis (tool)	augment	HCI	-	Mixed
Xiao et al. (2023)	Qual coding (deductive)	augment	HCI	-	Promising
Savelka et al. (2023)	Qual coding (deductive)	augment	NLP	-	Mixed
Zhang et al. (2025)	Qual coding (deductive)	augment	Qual-methods	0.46	Mixed
authors (2025)	Qual coding (deductive)	augment	Qual-methods	0.46	Mixed
Gao et al. (2024)	Qual coding (inductive)	augment	HCI	-	Promising
Dunivin (2024)	Qual coding (inductive)	replace	Qual-methods	-	Promising
Ngo et al. (2026)	Qual coding (inductive)	augment	HCI	-	Mixed
Wu et al. (2023)	Social-computing annotation	replace	CSCW	-	Mixed
Zhu et al. (2023)	Social-computing annotation	replace	CSCW	-	Mixed
Rajashekar et al. (2024)	Social-computing annotation	augment	HCI	-	Mixed
Argyle et al. (2023)	Synthetic participants	simulate	CSS	-	Mixed
Bisbee et al. (2023)	Synthetic participants	replace	CSS	-	Caution
Cheng et al. (2023)	Synthetic participants	simulate	NLP	-	Caution

Study	Method	Role	Community	Agree	Verdict
Wang et al. (2024)	Synthetic participants	replace	HCI	-	Caution
Bail (2024)	Synthetic participants	simulate	CSS	-	Mixed
Abbasiantaeb et al. (2024)	Synthetic participants	simulate	IR	-	Mixed
Huang et al. (2025)	Synthetic participants	simulate	CSS	-	Mixed
Salminen et al. (2025)	Synthetic participants	replace	HCI	-	Caution
De Paoli (2023)	Thematic analysis	augment	CSS	-	Mixed
Dai et al. (2023)	Thematic analysis	augment	NLP	-	Promising
Prescott et al. (2024)	Thematic analysis	augment	Health	-	Mixed
Nguyen-Trung (2025)	Thematic analysis	augment	Qual-methods	-	Mixed
Jain et al. (2025)	Thematic analysis	replace	Qual-methods	0.91	Promising
Castellanos et al. (2025)	Thematic analysis	augment	Health	0.80	Promising
Sharma et al. (2025)	Thematic analysis	augment	HCI	-	Promising
Wen et al. (2025)	Thematic analysis	augment	Qual-methods	-	Mixed
Kocaballi (2023)	UX/design critique	augment	HCI	-	Mixed
Duan et al. (2024)	UX/design critique	augment	HCI	-	Mixed
Duan et al. (2024)	UX/design critique	augment	HCI	-	Mixed
Duan et al. (2024)	UX/design critique	augment	HCI	-	Mixed
Guerino et al. (2025)	Usability/heuristic eval	replace	HCI	0.21	Caution
Zhong et al. (2025)	Usability/heuristic eval	replace	HCI	0.75	Mixed